TOPIC 2 • THEORY Advanced Mathematics • BSc Aerospace Engineering • UC3M

First-Order ODEs: Existence, Stability and Explicit Solutions

Pages 3 and 4 of the class notes, in their order: initial value problems and Picard's theorem, autonomous equations and the stability of their stationary points, and three methods that give explicit solutions (separation of variables, the integrating factor and exact equations), with every example of the notes solved step by step. Robinson formalises each step; material beyond the notes is labelled as a complement.

Ground truth: Class notes Notes.pdf, pp. 3–4 • Robinson (2004) Ch. 6 (pp. 38–45), Ch. 7 (pp. 46–58), Ch. 8 (pp. 59–74), Ch. 9 (pp. 75–88), Ch. 10 (pp. 89–98) • Lecture map ODE_lecture_theory_to_Robinson_chapters.txt , Part I (Sep 10, Sep 17)

≈ 55 min study time • 77 displayed equations • 11 figures

SECTION 01 • NOTES P. 3 • ROBINSON SEC. 6.1 • LECTURE SEP 10

1. Initial value problems

A differential equation alone has many solutions: $\dot{x} = \lambda x$ is solved by $Ae^{\lambda t}$ for every constant A. One extra piece of information, the state at one instant, selects one of them.

INITIAL VALUE PROBLEMS [NOTES, P. 3]

An **IVP** for a first-order ODE is

$$\begin{cases} y'(x) = f\big(x, y(x)\big), \\ y(x_0) = y_0. \end{cases}$$

We are looking for the **existence** of a solution as well as the **uniqueness** of the solution.

Reading note: the first line of p. 3 writes the initial condition as $y(a_0) = x_0$; the statement of Picard's theorem a few lines below writes $y(x_0) = y_0$, which is the form used here.

The notes use x as the independent variable and y as the unknown. Robinson writes the same problem with time t and unknown $x(t)$; both forms appear below and mean the same thing.

DEFINITION — INITIAL VALUE PROBLEM (IVP) [ROBINSON, SEC. 6.1, EQ. (6.2), P. 38]

An **initial value problem** is a differential equation together with an initial condition:

$$\frac{dx}{dt} = f(x,t), \qquad x(t_0) = x_0. \tag{6.2}$$

DEFINITION 6.1 — SOLUTION OF AN IVP [ROBINSON, P. 40]

Given an open interval I that contains t_0, a **solution** of (6.2) on I is a differentiable function $x(t)$ defined on I with $x(t_0) = x_0$ and $\dot{x}(t) = f(x(t), t)$ for all $t \in I$.

The definition deliberately allows a solution to live on a limited interval I rather than on all of $\mathbb{R}$. Section 3 shows that this is unavoidable even for very simple equations.

1.1 Complement (Robinson): the philosophy of a well-posed problem

LECTURE TERMINOLOGY (SEP 10) — HADAMARD WELL-POSEDNESS

A mathematical model is called **well posed** if

1. **a solution exists** (otherwise the model predicts nothing);
2. **the solution is unique** (otherwise the model cannot decide between futures);
3. **the solution depends continuously on the data** (a small error in the measured x_0 produces a small error in $x(t)$, at least over finite times).

The words "well-posed" and "Picard's theorem" are lecture terminology; Robinson never uses them. He discusses existence and uniqueness in Sec. 6.1 and states the result as Theorem 6.2 (lecture map, notes on differences).

Robinson gives the physical reading of uniqueness: if t is time and the initial condition is specified "now", a unique solution means the equation can be used to *predict the future*; uniqueness is equivalent to the model being **deterministic** [p. 39]. It also legitimises guessing: if a guessed function satisfies the IVP, it is *the* solution.

1.2 Complement (Robinson): existence is not automatic

AN IVP WITH NO SOLUTION (ROBINSON P. 38)

$$x^2 + t^2 \frac{dx}{dt} = 0, \qquad x(0) = c, \quad c \neq 0.$$

Evaluate the equation at $t = 0$: the second term vanishes and $x(0)^2 = 0$, i.e. $x(0) = 0$. This contradicts $x(0) = c \neq 0$, so no differentiable solution exists. The culprit is visible in the explicit form $\dot{x} = -x^2/t^2$: the right-hand side is not even defined at $t = 0$.

2. Picard's existence and uniqueness theorem

PICARD'S THEOREM FOR $y'(x) = f\big(x, y(x)\big)$ [NOTES, P. 3]

Let f and $\dfrac{\partial f}{\partial y}$ be continuous functions of x and y near the initial value (x_0, y_0), with $y_0 = y(x_0)$. Then there exists $h > 0$ such that the IVP

$$\begin{cases} y'(x) = f\big(x, y(x)\big), \\ y(x_0) = y_0 \end{cases}$$

has a **unique solution** for $|x - x_0| < h$.

Robinson states the same theorem with t as the independent variable and x as the unknown, so the derivative that must be continuous is $\partial f/\partial x$, and the interval $|x - x_0| < h$ of the notes becomes "some open interval I containing t_0".

THEOREM 6.2 (PICARD'S THEOREM IN THE LECTURE) [ROBINSON, P. 40]

If $f(x,t)$ and $\dfrac{\partial f}{\partial x}(x,t)$ are continuous for $a < x < b$ and $c < t < d$, then for any $x_0 \in (a,b)$ and $t_0 \in (c,d)$ the initial value problem

$$\frac{dx}{dt} = f(x,t), \qquad x(t_0) = x_0$$

has a unique solution on some open interval I containing t_0.

Three features deserve attention:

- **The hypothesis is easy to check:** continuity of f and of one partial derivative, in a rectangle around (x_0, t_0).
- **The derivative is with respect to the unknown** x, not t. Nothing is required of $\partial f/\partial t$.
- **The conclusion is local:** it says nothing about how large I is (Robinson Fig. 6.2). Section 3 shows that I can indeed be short.

2.1 Complement (Robinson): the Lipschitz condition

DEFINITION — LIPSCHITZ CONTINUITY [ROBINSON, FOOTNOTE TO THM 6.2, EQ. (6.5), P. 40]

f is **Lipschitz continuous in** x if there is a constant L such that

$$|f(x,t)-f(y,t)| \le L\,|x-y| \tag{6.5}$$

for all x, y in the region considered. Robinson notes that this weaker hypothesis already suffices for Theorem 6.2.

PROPOSITION — A BOUNDED DERIVATIVE IMPLIES LIPSCHITZ [ROBINSON, EXERCISE 6.2, P. 44]

If f is differentiable in x with $\left|\dfrac{\partial f}{\partial x}\right| \le L$ for $a \le x \le b$, then $|f(x,t)-f(y,t)| \le L|x-y|$ there.

PROOF (MEAN VALUE THEOREM)

Fix t and $a \le y < x \le b$. By the Mean Value Theorem there is $\xi \in (y,x)$ with $f(x,t)-f(y,t) = (x-y)\,\dfrac{\partial f}{\partial x}(\xi,t)$. Taking absolute values and using the bound, $|f(x,t)-f(y,t)| = |x-y|\,\left|\dfrac{\partial f}{\partial x}(\xi,t)\right| \le L\,|x-y|$. The case $x=y$ is trivial, and $x<y$ follows by symmetry.

■

A continuous function on a closed bounded rectangle is bounded, so the hypothesis of Theorem 6.2 gives $|\partial f/\partial x| \le L$ on any closed sub-rectangle, and hence a Lipschitz bound. The converse is false: $f(x)=|x|$ is Lipschitz with $L=1$ (because $\big||x|-|y|\big| \le |x-y|$) but not differentiable at 0.

2.2 Complement (Robinson): proof of uniqueness and of continuous dependence

Robinson leaves the proofs as guided exercises. The uniqueness argument (Exercise 6.3) is short, and the same estimate also delivers the third Hadamard condition.

PROPOSITION — STABILITY ESTIMATE

Suppose $|f(x,t)-f(y,t)| \le L|x-y|$ and let $x(t), y(t)$ solve $\dot{x} = f(x,t)$ with $x(t_0)=x_0, y(t_0)=y_0$. Then for $t \ge t_0$

$$|x(t)-y(t)| \le |x_0-y_0|\,e^{L(t-t_0)}.$$

In particular, if $x_0 = y_0$ then $x(t)=y(t)$: **the solution is unique**.

PROOF (ROBINSON EXERCISES 6.3 AND 9.7)

1. Let $z(t) = x(t)-y(t)$. Subtracting the two equations: $\dot{z} = f(x(t),t) - f(y(t),t)$.
2. Let $Z(t) = z(t)^2 = |z|^2$. By the chain rule with outer function $u \mapsto u^2$:

$$\frac{dZ}{dt} = 2z\,\frac{dz}{dt} = 2z\,\big[f(x,t)-f(y,t)\big].$$

3. Bound it with the Lipschitz condition: $\dfrac{dZ}{dt} \le 2|z|\,|f(x,t)-f(y,t)| \le 2|z|\cdot L|z| = 2L\,Z$.

4. Multiply by the positive factor e^{-2Lt} and use the product rule:

$$\frac{d}{dt}\Big(e^{-2Lt}Z\Big) = e^{-2Lt}\dot Z - 2Le^{-2Lt}Z = e^{-2Lt}\big(\dot Z - 2LZ\big) \le 0.$$

5. A function with non-positive derivative is non-increasing, so for $t \ge t_0$: $e^{-2Lt}Z(t) \le e^{-2Lt_0}Z(t_0)$, i.e. $Z(t) \le Z(t_0)\,e^{2L(t-t_0)}$. Taking square roots: $|z(t)| \le |z(t_0)|\,e^{L(t-t_0)}$.

6. If $x_0 = y_0$ then $z(t_0) = 0$, so $|z(t)| \le 0$ and $x(t) = y(t)$ for all $t \ge t_0$.

■

Interpretation. Two solutions that start ε apart remain within $\varepsilon e^{L(t-t_0)}$ of each other: the error can grow, but at most exponentially, and it tends to zero with ε on any finite time interval. This is exactly continuous dependence on the initial data. For $\dot x = \lambda x$ with $\lambda \ge 0$ (so $L = \lambda$) the bound is attained: $|x(t)-y(t)| = |x_0 - y_0|e^{\lambda(t-t_0)}$; for $\lambda < 0$ the solutions actually approach each other.

2.3 Complement (Robinson): idea of the existence proof, Picard iteration

Existence is harder than uniqueness. Robinson's Exercise 6.4 sketches it for $\dot x = f(x), x(0) = x_0$ [pp. 44-45].

1. **Integral form.** Integrating the ODE from 0 to t (Barrow: $\int_0^t \dot x\,d\tilde t = x(t) - x(0)$) gives the equivalent integral equation

$$x(t) = x_0 + \int_0^t f\big(x(\tilde t)\big)\,d\tilde t. \tag{E6.5}$$

   Conversely, differentiating (E6.5) with the Fundamental Theorem of Calculus returns $\dot x = f(x)$, and $t = 0$ returns $x(0) = x_0$.

2. **Iteration.** Start from the constant guess $x_0(t) = x_0$ and improve it repeatedly:

$$x_{n+1}(t) = x_0 + \int_0^t f\big(x_n(\tilde t)\big)\,d\tilde t. \tag{E6.6}$$

3. **Contraction.** Using (E6.4) $|f(x)-f(y)| \le L|x-y|$, for $0 \le t \le 1/(2L)$:

$$|x_{n+1}(t) - x_n(t)| \le \int_0^t L\,|x_n - x_{n-1}|\,d\tilde t \le L\,t\,\max|x_n - x_{n-1}| \le \tfrac{1}{2}\max|x_n - x_{n-1}|. \tag{E6.7}$$

   Each step halves the maximum difference, so the differences shrink like 2^{-n} and $x_n(t)$ converges to some $x_\infty(t)$.

4 **Limit.** Letting $n \to \infty$ in (E6.6) shows that x_∞ satisfies (E6.5), so it solves the IVP; the previous proposition shows it is the only one.

ILLUSTRATION: PICARD ITERATES FOR $\dot{x} = x$, $x(0) = 1$

Here $f(x) = x$ and $L = 1$. Each iterate integrates the previous one:

$$x_1 = 1 + \int_0^t 1 \, d\tilde{t} = 1 + t, \qquad x_2 = 1 + \int_0^t (1 + \tilde{t}) \, d\tilde{t} = 1 + \Big[\tilde{t} + \tfrac{\tilde{t}^2}{2}\Big]_0^t = 1 + t + \frac{t^2}{2},$$

$$x_3 = 1 + \Big[\tilde{t} + \tfrac{\tilde{t}^2}{2} + \tfrac{\tilde{t}^3}{6}\Big]_0^t = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}, \qquad \dots, \qquad x_n = \sum_{j=0}^{n} \frac{t^j}{j!} \longrightarrow e^t.$$

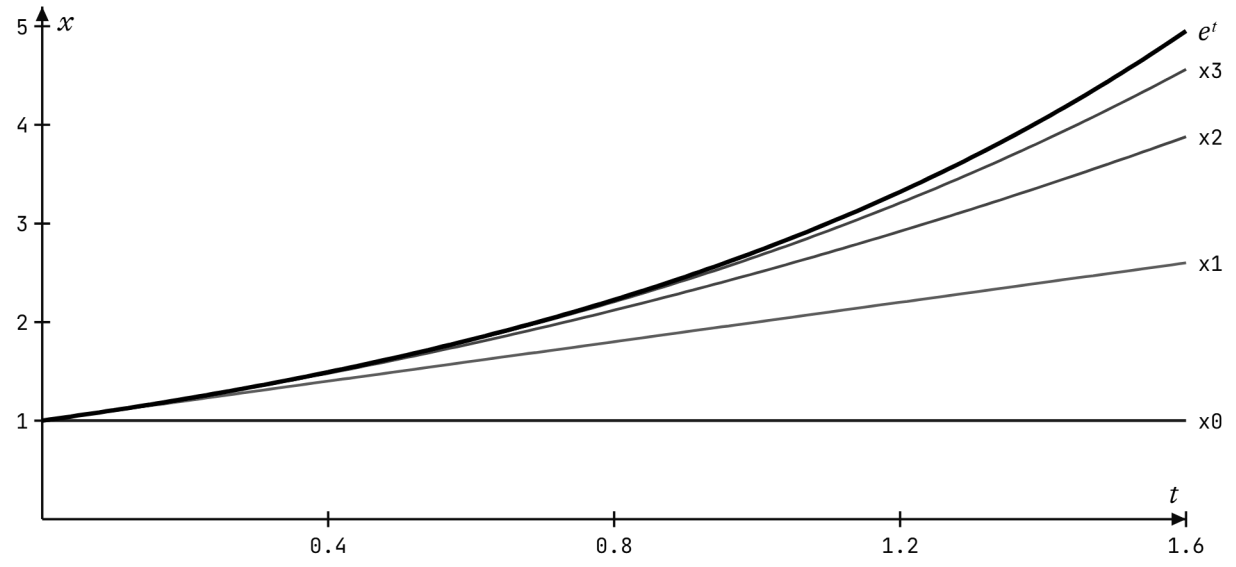


Figure 2.1. *The first Picard iterates $x_0 = 1$, $x_1 = 1 + t$, $x_2 = 1 + t + t^2/2$, x_3 for $\dot{x} = x$, $x(0) = 1$, approaching the exact solution e^t (indigo). Each iterate is the Taylor polynomial of e^t.*

SECTION 03 • ROBINSON SEC. 6.1, 6.3, 6.4, EX. 6.1 • COMPLEMENT

3. Complement: when the theorem fails or stops short

The class notes state Picard's theorem without counterexamples. Robinson's examples show why each hypothesis is there and why the conclusion is only local.

3.1 Loss of uniqueness: $\dot{x} = \sqrt{x}$

Robinson's example [Eq. (6.3), p. 39]:

$$\frac{dx}{dt} = \sqrt{x}, \qquad x(0) = 0. \tag{6.3}$$

The obvious solution is $x(t) \equiv 0$. But for any $c > 0$ the function

$$x_c(t) = \begin{cases} 0, & t \le c, \\ \dfrac{(t-c)^2}{4}, & t > c \end{cases}$$

is also a solution. **Check** for $t > c$ (chain rule with inner function $t - c$): $\dot{x}_c = \dfrac{2(t-c)}{4} \cdot 1 = \dfrac{t-c}{2}$, while $\sqrt{x_c} = \sqrt{(t-c)^2/4} = \dfrac{|t-c|}{2} = \dfrac{t-c}{2}$. At $t = c$ both one-sided derivatives are 0, so x_c is differentiable there. The solution "waits" at 0 and then "decides" at the arbitrary time c to leave [p. 39]: the model is not deterministic.

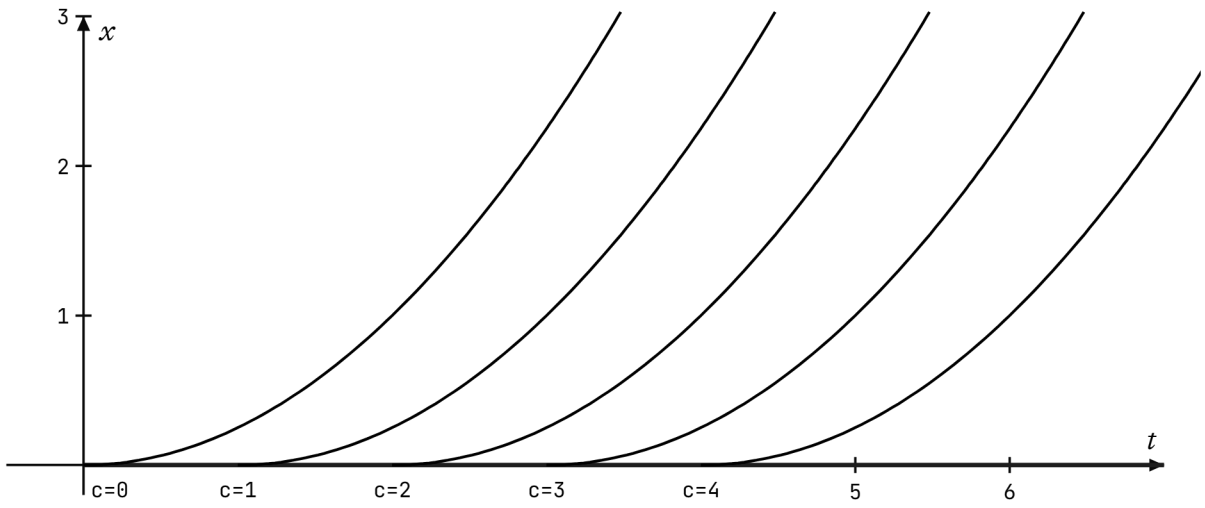


Figure 2.2. *Infinitely many solutions of $\dot{x} = \sqrt{x}$, $x(0) = 0$ (Robinson (6.3), Fig. 6.1, p. 39): the zero solution (red) and the curves $x_c(t)$ that stay at 0 until $t = c$ and then follow $(t-c)^2/4$.*

Why Theorem 6.2 does not apply. $f(x) = x^{1/2}$ is continuous, but $f'(x) = \tfrac{1}{2}x^{-1/2} \to \infty$ as $x \downarrow 0$, so $\partial f/\partial x$ is not continuous at the initial point $x_0 = 0$ [p. 40; Fig. 6.3, p. 41]. Nor is f Lipschitz near 0: $\dfrac{|f(x) - f(0)|}{|x - 0|} = \dfrac{\sqrt{x}}{x} = \dfrac{1}{\sqrt{x}} \to \infty$ as $x \downarrow 0$, so no constant L works. Note the logic: the theorem gives a *sufficient* condition; its failure here is consistent with the non-uniqueness, not a proof of it – the explicit family x_c is the proof.

THE SAME PHENOMENON FOR $\dot{x} = x^{1/3}$ (ROBINSON EXERCISE 6.1 (III))

With $x(0) = 0$, $x \equiv 0$ is a solution. Separating the variables for $x > 0$: $x^{-1/3}dx = dt$. An antiderivative of $x^{-1/3}$ is $\frac{3}{2}x^{2/3}$, so integrating from the time c at which x leaves 0:

$$\left[\tfrac{3}{2}x^{2/3}\right]_0^{x(t)} = \left[\tilde{t}\right]_c^t \implies \tfrac{3}{2}x(t)^{2/3} = t - c \implies x(t) = \left(\frac{2(t-c)}{3}\right)^{3/2}, \quad t \geq c.$$

Check: $\dot{x} = \frac{3}{2}\left(\frac{2(t-c)}{3}\right)^{1/2} \cdot \frac{2}{3} = \left(\frac{2(t-c)}{3}\right)^{1/2} = x^{1/3}$. Again infinitely many solutions (and their mirror images $-x$), because $f'(x) = \frac{1}{3}x^{-2/3}$ blows up at $x = 0$.

3.2 Finite-time blow-up: $\dot{x} = x^2$

Robinson's second cautionary example [Sec. 6.3, Eq. (6.6), p. 41]:

$$\frac{dx}{dt} = x^2, \qquad x(0) = x_0. \tag{6.6}$$

Here $f(x) = x^2$ and $f'(x) = 2x$ are continuous everywhere, so Theorem 6.2 guarantees a unique solution near $t = 0$ for every x_0. For $x_0 \neq 0$ the solution is [Eq. (6.7), p. 42]

$$x(t) = \frac{1}{x_0^{-1} - t}. \tag{6.7}$$

Verification. At $t = 0$, $x = 1/x_0^{-1} = x_0$. Write $x = u^{-1}$ with $u(t) = x_0^{-1} - t$, $du/dt = -1$. Chain rule:

$$\frac{dx}{dt} = \frac{d}{du}\left(u^{-1}\right) \cdot \frac{du}{dt} = \left(-u^{-2}\right) \cdot (-1) = \frac{1}{(x_0^{-1} - t)^2} = x(t)^2 \ \checkmark.$$

By uniqueness this is *the* solution. (Section 6.2 derives it by separation of variables, Robinson Example 8.1.)

- $x_0 > 0$**:** the denominator $x_0^{-1} - t$ decreases to 0 at $t^* = 1/x_0$, so $x(t) \to +\infty$ as $t \uparrow 1/x_0$: the solution **blows up in finite time**. Backwards in time it decays to 0. Maximal interval of existence: $(-\infty, 1/x_0)$.
- $x_0 < 0$**:** the blow-up happens in the past, $x \to -\infty$ as $t \downarrow 1/x_0 < 0$. Maximal interval: $(1/x_0, +\infty)$.
- $x_0 = 0$**:** $x \equiv 0$, defined for all $t \in \mathbb{R}$.

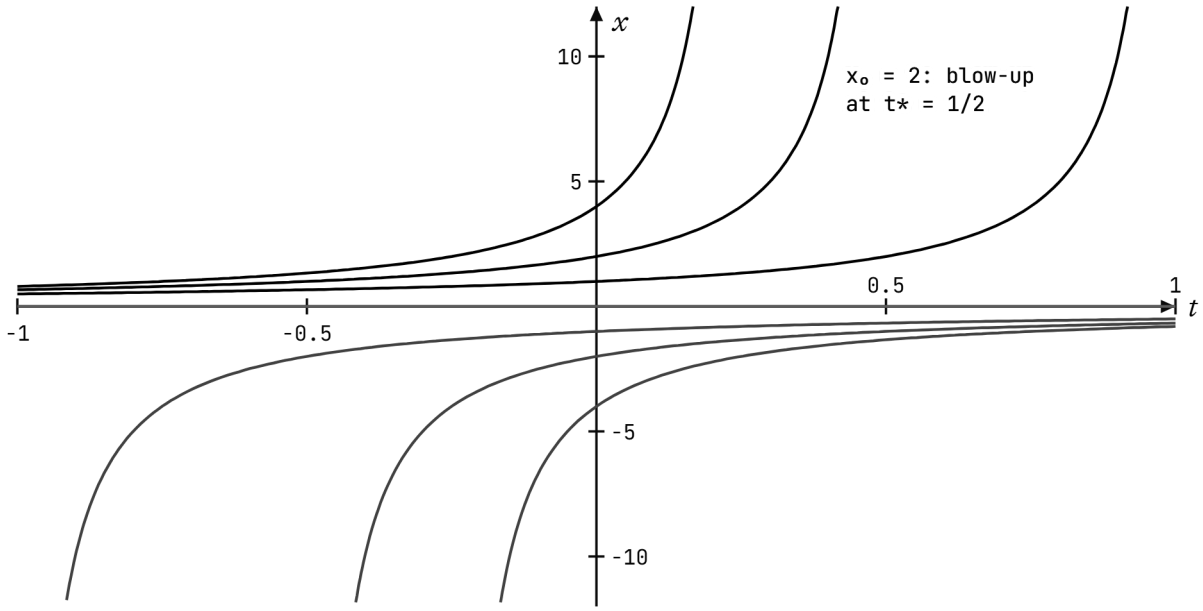


Figure 2.3. *Solutions $x(t) = 1/(x_0^{-1} - t)$ of $\dot{x} = x^2$ for $x_0 = 1, 2, 4$ (indigo) and $x_0 = -1, -2, -4$ (green), cf. Robinson Fig. 6.4, p. 43. Positive data blow up at $t^* = 1/x_0 > 0$; negative data blow up in the past at $t^* = 1/x_0 < 0$; only $x_0 = 0$ gives a solution on all of $\mathbb{R}$.*

WHY THE INTERVAL IN THEOREM 6.2 MUST BE "SOME INTERVAL"

$f(x) = x^2$ is as smooth as a function can be, yet solutions do not exist for all time. The reason is that f is Lipschitz on every bounded interval but not on all of $\mathbb{R}$: $|x^2 - y^2| = |x + y|\,|x - y|$ and $|x + y|$ is unbounded. Fast growth ($\dot{x} \sim x^2$) feeds on itself. Compare $\dot{x} = |x|$, whose solutions $x_0 e^t$ (for $x_0 > 0$) exist forever although its phase line looks identical (Robinson Sec. 7.5.3).

3.3 Summary of the failure modes

EQUATION	HYPOTHESES OF THM 6.2	WHAT HAPPENS	LESSON
$x^2 + t^2\dot{x} = 0,\ x(0) = c \neq 0$	$f = -x^2/t^2$ undefined at $t = 0$	No solution	Existence can fail
$\dot{x} = \sqrt{x},\ x(0) = 0$	$\partial f/\partial x$ infinite at $x = 0$	Infinitely many solutions	Uniqueness needs Lipschitz
$\dot{x} = x^{1/3},\ x(0) = 0$	$\partial f/\partial x$ infinite at $x = 0$	Infinitely many solutions	Same as above
$\dot{x} = x^2,\ x(0) = x_0 > 0$	Satisfied everywhere	Unique, but blows up at $t = 1/x_0$	The theorem is only local

A ONE-MILLION-DOLLAR OPEN PROBLEM (ROBINSON SEC. 6.4, PP. 42–43)

It is still not known whether the three-dimensional Navier–Stokes equations have unique solutions that exist for all positive times. Every CFD computation of a wing or an engine assumes they do, yet one cannot prove that a numerical approximation is close to the "true solution" without knowing that such a solution exists. The question is one of the Clay Mathematics Institute's seven Millennium Prize Problems: prove global existence and uniqueness, or find one initial condition for which the solution breaks down.

SECTION 04 • NOTES P. 3 • ROBINSON SEC. 7.1–7.3, 7.5 • LECTURE SEP 10

4. Autonomous first-order ODEs and stationary points

4.1 Autonomous equations

AUTONOMOUS FIRST-ORDER ODES [NOTES, P. 3]

A first-order ODE $x'(t) = f$ is **autonomous** when f only depends on the dependent variable x:

- $x'(t) = \big(x(t)\big)^3$ is autonomous, because $f(x) = x^3$ does not depend directly on t;
- $x'(t) = t^2\, x(t)$ is **not** autonomous, because $f(x, t) = t^2 x$.

For a general first-order equation we will look for explicit formulas. For the special class of **autonomous** equations

$$\frac{dx}{dt} = f(x) \tag{7.1}$$

there is a faster and completely rigorous alternative: describe *how* the solutions behave ("every solution starting between 0 and 1 tends to 1") without writing them down. The method only needs the sign of f [Robinson, Ch. 7 intro, p. 46].

4.2 Stationary points and their stability

STATIONARY POINTS [NOTES, P. 3]

Stationary points are the points x where $f(x) = 0$.

- If $f(x) = 0$ and $f'(x) > 0$, then x is **unstable**.
- If $f(x) = 0$ and $f'(x) < 0$, then x is a **stable** stationary point.

A stationary point x^* gives a constant solution: if $x(0) = x^*$, then $x'(0) = f(x^*) = 0$ and, by uniqueness (Section 2), $x(t) \equiv x^*$ for all t. Robinson's treatment makes each part of the criterion precise.

By Theorem 6.2 (Section 2), if f and f' are continuous, a solution of (7.1) is completely determined by its value at any one time. Think of $x(t)$ as the position of a particle moving on a line, with velocity $\dot{x} = f(x)$ that depends only on where the particle is [Sec. 7.1, p. 46].

DEFINITION — STATIONARY POINT [ROBINSON, P. 47]

A **stationary point** of (7.1) is a value x^* with $f(x^*) = 0$. The constant function $x(t) \equiv x^*$ is then a solution, since $\dot{x} = 0 = f(x^*)$. (Other names: equilibrium point, critical point, fixed point; Robinson reserves the last two for other meanings.)

- Where $f(x) > 0, \dot{x} > 0$: $x(t)$ increases and the particle moves **right**.
- Where $f(x) < 0, \dot{x} < 0$: $x(t)$ decreases and the particle moves **left**.

PROPOSITION — SOLUTIONS ARE MONOTONE AND NEVER REACH A STATIONARY POINT IN FINITE TIME

If f and f' are continuous, a non-constant solution of $\dot{x} = f(x)$ cannot change direction and cannot cross or reach a stationary point.

PROOF [ROBINSON P. 47; EXERCISE 7.8]

Suppose the solution reversed direction at some time t^*. Then it is instantaneously at rest, $\dot{x}(t^*) = f(x(t^*)) = 0$, so $x^* = x(t^*)$ is a stationary point. But $x(t) \equiv x^*$ is also a solution through the same point at the same time, and by uniqueness the two solutions coincide: the particle never moved, a contradiction. The same argument rules out reaching a stationary point at any finite time. So $\dot{x} = f(x(t))$ never vanishes, and by continuity it keeps one sign: $x(t)$ is strictly monotone.

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DEFINITIONS — STABLE, ATTRACTING, UNSTABLE [ROBINSON, SEC. 7.2, EQS. (7.2)-(7.3), P. 48]

- x^* is **stable** if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|x_0 - x^*| < \delta \implies |x(t) - x^*| < \varepsilon \quad \text{for all } t \geq 0. \tag{7.2}$$

  "Start close enough and you stay close."

- x^* is **attracting** if there is a $\delta > 0$ such that

$$|x_0 - x^*| < \delta \implies x(t) \to x^* \text{ as } t \to +\infty. \tag{7.3}$$

- x^* is **unstable** if it is not stable: there is an $\varepsilon > 0$ such that for every $\delta > 0$ some x_0 with $|x_0 - x^*| < \delta$ gives $|x(t) - x^*| > \varepsilon$ for some $t > 0$.

In one dimension attracting implies stable (Exercise 7.9), but stable does not imply attracting. Robinson's counterexample is $\dot{x} = -x$ for $x < 0$, $\dot{x} = 0$ for $0 \leq x \leq 1$, $\dot{x} = 1 - x$ for $x > 1$: every point of $[0, 1]$ is stationary and stable, but none is attracting, because nearby points in $[0, 1]$ do not move at all [p. 48; Fig. 7.3, p. 49].

STABILITY CRITERION [ROBINSON, SEC. 7.3, P. 49, FIG. 7.5]

Let $f(x^*) = 0$ with f continuously differentiable.

- If $f'(x^*) < 0$, then x^* is **stable** (in fact attracting).
- If $f'(x^*) > 0$, then x^* is **unstable**.
- If $f'(x^*) = 0$, the test is inconclusive: the point may be stable, unstable, or **semi-stable** (stable on one side, unstable on the other) [Fig. 7.6, p. 50].

Why it works (graphical argument, as in the book). If $f'(x^*) < 0$, the graph of f crosses zero going downwards: $f > 0$ just to the left of x^* and $f < 0$ just to the right. Both arrows point towards x^*. If $f'(x^*) > 0$ the signs are reversed and both arrows point away.

Why it works (linearisation – complementary derivation). Write $x(t) = x^* + \xi(t)$ with a small perturbation ξ. Since x^* is constant, $\dot{x} = \dot{\xi}$. Taylor's theorem gives $f(x^* + \xi) = f(x^*) + f'(x^*)\,\xi + O(\xi^2) = f'(x^*)\,\xi + O(\xi^2)$. Dropping the quadratic term,

$$\dot{\xi} \approx f'(x^*)\,\xi \quad \Longrightarrow \quad \xi(t) \approx \xi(0)\, e^{f'(x^*)\,t}.$$

The perturbation decays if $f'(x^*) < 0$ and grows if $f'(x^*) > 0$. The quantity $1/|f'(x^*)|$ is the time scale of the approach or departure. When $f'(x^*) = 0$ the linear term vanishes and the neglected $O(\xi^2)$ terms decide, which is why the test is then silent.

4.3 Reading the phase line: the sketch of the notes

The notes draw a function f whose graph is an arch (a downward parabola) with two zeros $\underline{x} < \overline{x}$: $f > 0$ between them and $f < 0$ outside [Notes, p. 3]. For $x'(t) = f\big(x(t)\big)$:

- If $x_0 = \underline{x}$, then $x'(0) = f(\underline{x}) = 0$; if $x_0 = \overline{x}$, then $x'(0) = f(\overline{x}) = 0$. The solution stays where it started.
- If $\underline{x} < x_0 < \overline{x}$, then $f(x_0) > 0$, so $x' > 0$: the solution moves **right**.
- If $x_0 < \underline{x}$ or $x_0 > \overline{x}$, then $f(x_0) < 0$, so $x' < 0$: the solution moves **left**.

Both arrows next to $\underline{x}$ point away from it: point 1 is **unstable**, since $f'(\underline{x}) > 0$ (the arch goes up through its left zero). Both arrows next to $\overline{x}$ point towards it: point 2 is **stable**, since $f'(\overline{x}) < 0$. For a concrete arch $f(x) = (x - \underline{x})(\overline{x} - x)$, the product rule gives $f'(x) = (\overline{x} - x) - (x - \underline{x})$, so $f'(\underline{x}) = \overline{x} - \underline{x} > 0$ and $f'(\overline{x}) = -(\overline{x} - \underline{x}) < 0$, as the picture says.

The same procedure works for any f (Robinson's recipe):

1. Sketch the graph of $f(x)$ against x (only its sign matters).
2. Mark the zeros x^* of f on a horizontal line: these are the stationary points.
3. On each interval between stationary points draw an arrow to the right if $f > 0$ there and to the left if $f < 0$.
4. Read off the behaviour: a solution moves in the direction of the arrow and, being monotone, either tends to the next stationary point or to $\pm\infty$ (Robinson Exercise 7.10).

The resulting picture is the **phase diagram** (or **phase portrait**) [p. 47, Figs. 7.1-7.2].

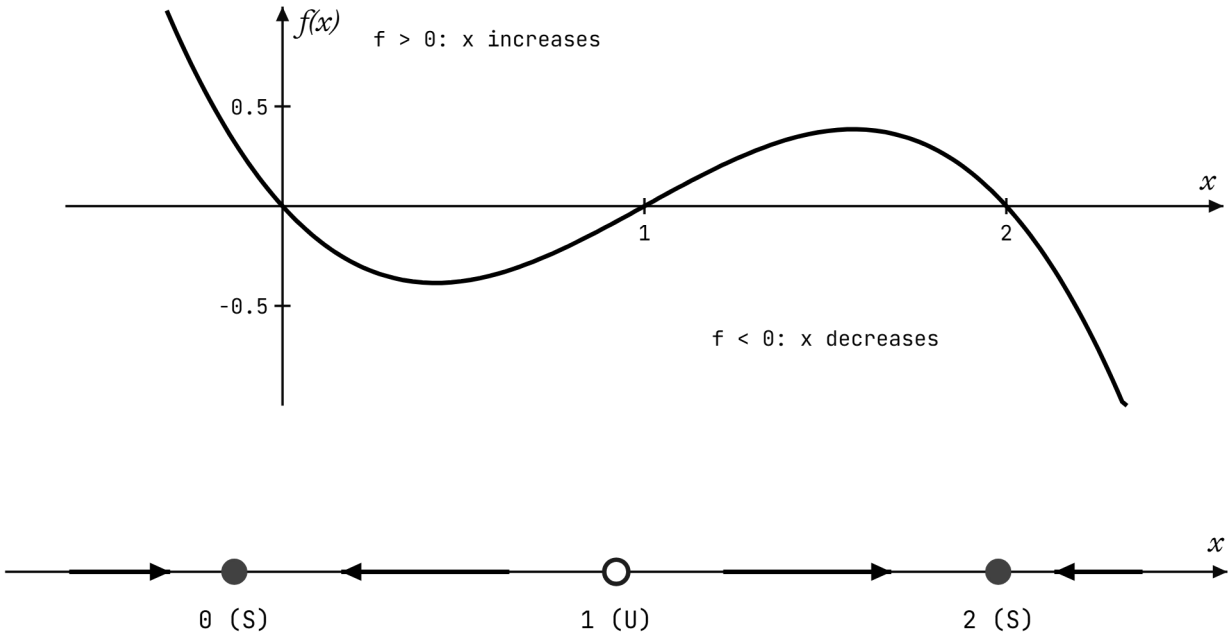


Figure 2.4. *Building a phase diagram (method of Robinson Figs. 7.1–7.2) for $\dot{x} = -x(1-x)(2-x)$, Exercise 7.1 (iv). Top: the graph of f. Bottom: the phase line. Arrows point right where $f > 0$ and left where $f < 0$; filled points are stable (S), hollow points unstable (U).*

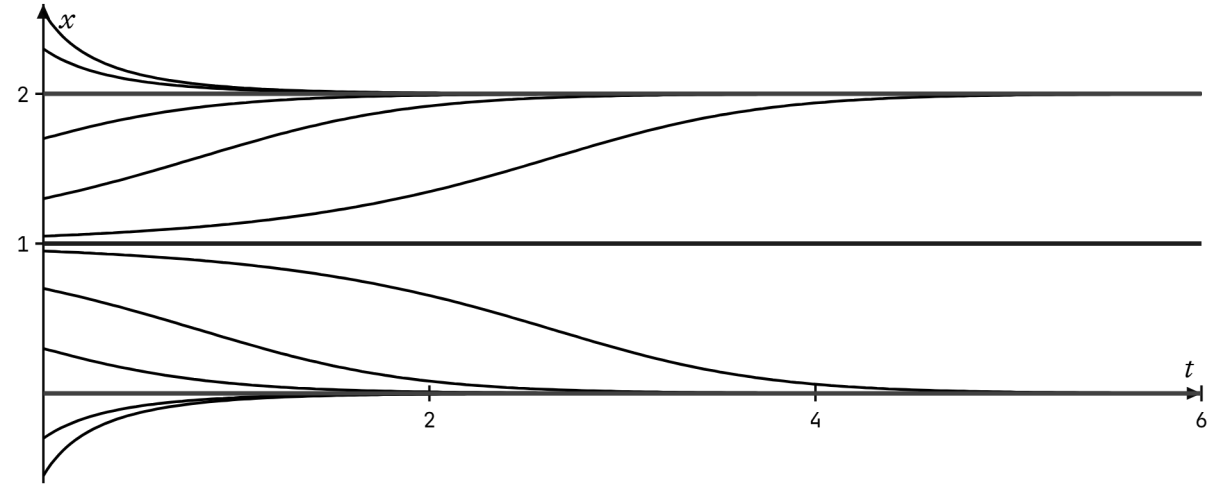


Figure 2.5. *Solutions $x(t)$ of the same equation from several initial values (numerical integration). They are monotone, never cross the constant solutions $x = 0, 1, 2$ and approach the stable ones; compare Robinson Fig. 7.8.*

EXAMPLE — THE CUBIC OF FIGURE 2.4

$f(x) = -x(1-x)(2-x) = -(x^3 - 3x^2 + 2x)$, so $f'(x) = -(3x^2 - 6x + 2)$. Then $f'(0) = -2 < 0$ (stable), $f'(1) = -(3 - 6 + 2) = 1 > 0$ (unstable), $f'(2) = -(12 - 12 + 2) = -2 < 0$ (stable), in agreement with the arrows.

4.4 Concrete example: the logistic equation

CONCRETE EXAMPLE [NOTES, P. 3]

With constants $k, M > 0$:

$$\begin{cases} x'(t) = kx\left(1 - \dfrac{x}{M}\right) = kx - \dfrac{k}{M}x^2 & \text{(non-linear)}, \\ x(0) = x_0 \geq 0. \end{cases}$$

This is the logistic correction of the population model of Topic 1, Section 5.4: for $x \ll M$ the term $\frac{k}{M}x^2$ is negligible and $x' \approx kx$; as x approaches M the growth rate falls to zero [Robinson, Sec. 8.5, Eq. (8.12), p. 66].

1. **Critical points.** $f(x) = kx(1 - x/M) = 0$ gives $x = 0$ and $x = M$.
2. **Derivative.** From $f(x) = kx - \frac{k}{M}x^2$: $f'(x) = k - \dfrac{2k}{M}x$.
3. **At $x = 0$: $f'(0) = k - 0 = k > 0$,** so $x = 0$ is **unstable**.
4. **At $x = M$: $f'(M) = k - 2k = -k < 0$,** so $x = M$ is **stable**.

The notes sketch the solution curves $x(t)$ against t: each curve is a solution of the ODE. Point 0 is unstable because a slight variation goes to M; if $x_0 = 0$, x stays at 0; if $x_0 \neq 0$, x ends up at M [Notes, p. 3]. Near M the linearisation of Section 4.2 gives $\xi(t) \approx \xi(0)e^{-kt}$: the population relaxes to M on the time scale $1/k$. The phase line alone cannot give k or M from data; that needs the explicit solution of Section 6.3.

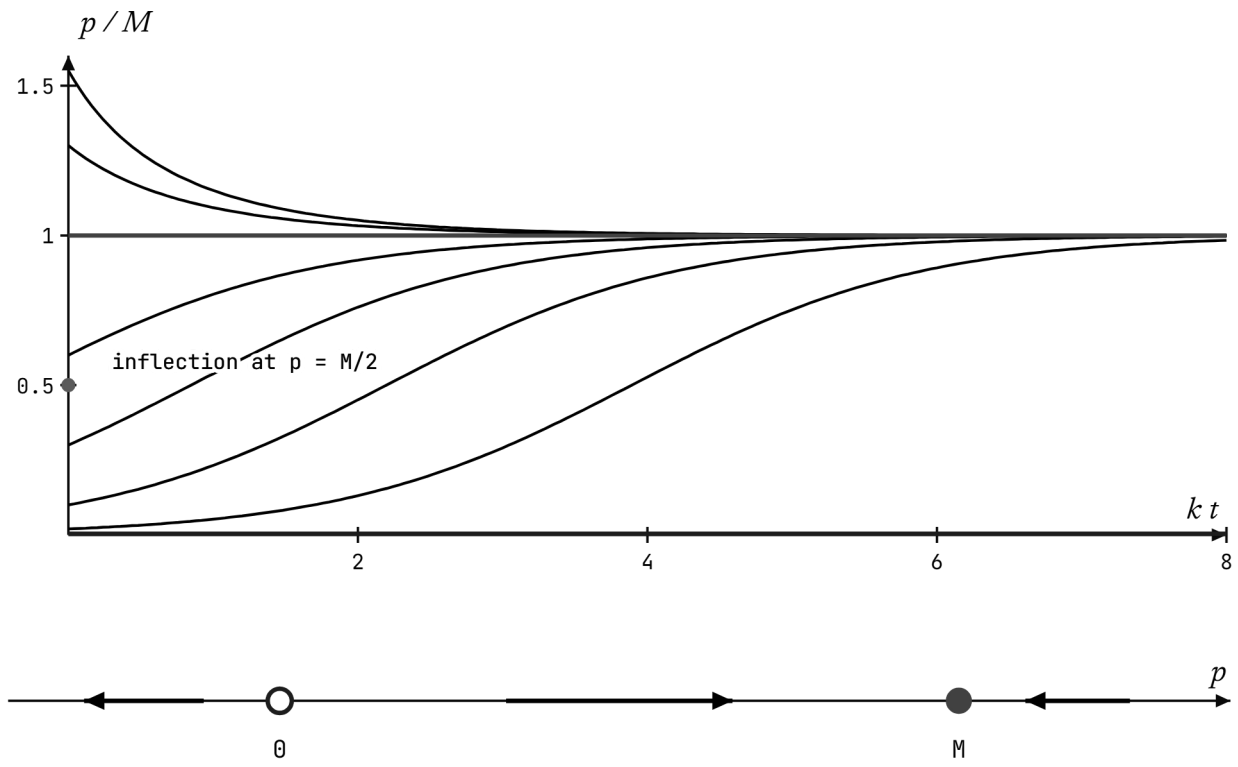


Figure 2.6. *Top: solutions of the logistic equation (8.12) in units p/M and kt, for several initial values. Bottom: the phase line (Robinson Fig. 8.3, p. 66), with the unstable equilibrium $p = 0$ (hollow) and the stable equilibrium $p = M$ (filled).*

EXAMPLE — TERMINAL VELOCITY (ROBINSON SEC. 7.5.2, EQ. (7.5), P. 52)

For downward speed $v > 0$: $\dot{v} = f(v) = g - (k/m)v^2$. The unique positive stationary point is $v^* = \sqrt{mg/k}$, and $f'(v) = -2kv/m$, so $f'(v^*) = -2\sqrt{gk/m} < 0$: terminal velocity is stable and attracts every initial speed. The explicit $\tanh$ solution is derived in Topic 1, Section 6.5.

WHAT THE PHASE LINE CANNOT TELL YOU (ROBINSON SEC. 7.5.3, P. 53)

$\dot{x} = |x|$ and $\dot{x} = x^2$ have the same phase diagram (one semi-stable point at 0, arrows pointing right everywhere). But for $x_0 > 0$ the first gives $x = x_0 e^t$, defined for all time, while the second gives $x = 1/(x_0^{-1} - t)$, which blows up at $t = 1/x_0$. The phase line captures the direction of motion, not its speed.

SECTION 05 • ROBINSON SEC. 7.4, 7.6, EX. 7.5-7.6 • COMPLEMENT

5. Complement: structural stability and bifurcations

Models contain parameters (a growth rate, a load, a drag coefficient) that are never known exactly. It matters whether a small change in a parameter changes the picture qualitatively.

DEFINITIONS [ROBINSON, SEC. 7.4, P. 50]

If a small change to f changes the phase diagram drastically, the equation has undergone a **bifurcation**. If no small change to f can change the qualitative phase diagram, $\dot{x} = f(x)$ is **structurally stable**.

If $f'(x^*) \neq 0$, a small change to f (for instance adding a small constant, which shifts the graph up or down) still leaves a nearby zero with the same sign of slope, hence a nearby stationary point of the same type (Exercise 7.11). So **a bifurcation near x^* requires $f'(x^*) = 0$** – exactly the degenerate case of the derivative test.

5.1 The pitchfork bifurcation

Robinson's worked example [Sec. 7.6, Eq. (7.6), pp. 54–56]:

$$\dot{x} = x\,(k - x^2), \qquad f(x) = kx - x^3, \qquad f'(x) = k - 3x^2.$$

- $k < 0$**:** $k - x^2 < 0$ for every x, so the only stationary point is $x^* = 0$, with $f'(0) = k < 0$: stable.
- $k = 0$**:** $f(x) = -x^3$ and $f'(0) = 0$ – the test fails. The sign of f (> 0 for $x < 0$, < 0 for $x > 0$) shows 0 is still stable. The vanishing derivative signals a possible bifurcation.
- $k > 0$**:** three stationary points $x^* = 0, \pm\sqrt{k}$. $f'(0) = k > 0$: unstable. $f'(\pm\sqrt{k}) = k - 3k = -2k < 0$: both stable.

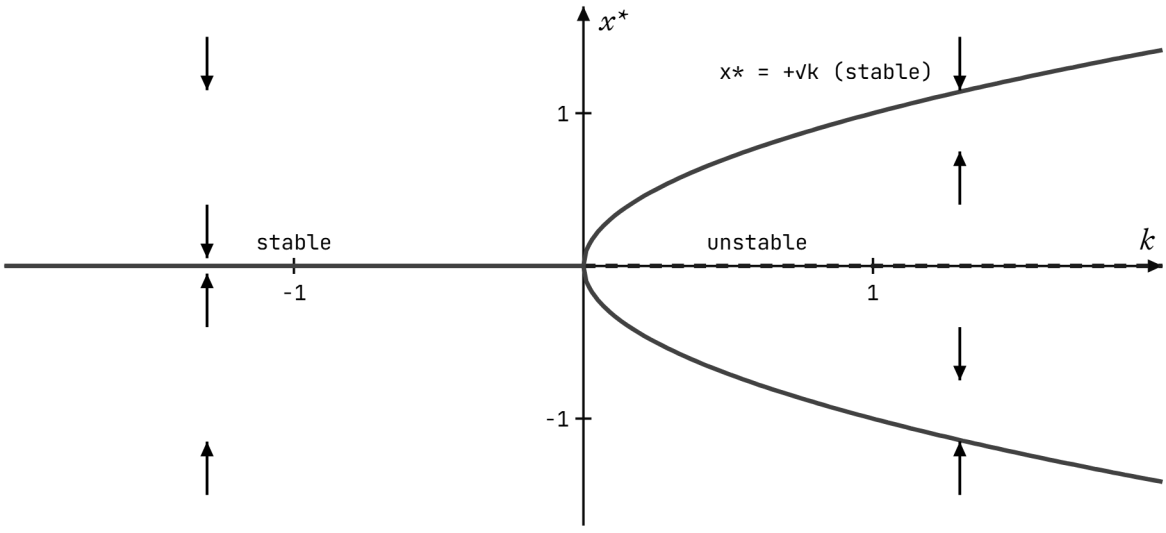


Figure 2.7. *Bifurcation diagram of the pitchfork $\dot{x} = x(k - x^2)$ (Robinson Sec. 7.6, Fig. 7.14). Solid green: stable stationary points; dashed red: unstable. At $k = 0$ the single stable point $x^* = 0$ splits into three.*

PHYSICAL READING (COMPLEMENTARY)

The pitchfork is the generic picture of **symmetric buckling**: a slender strut under axial load stays straight ($x^* = 0$) below the critical load and, above it, the straight state becomes unstable and the strut deflects to one of two mirror-image positions $\pm\sqrt{k}$. Which one is chosen depends on imperfections – the same sensitivity that makes $k = 0$ structurally unstable.

5.2 Saddle-node and transcritical bifurcations (Exercises 7.5 and 7.6)

- **Saddle-node:** $\dot{x} = x^2 - k$. For $k < 0$ there are no stationary points; for $k > 0$ there are two, $x^* = \pm\sqrt{k}$, with $f'(x) = 2x$: $-\sqrt{k}$ stable ($f' = -2\sqrt{k}$), $+\sqrt{k}$ unstable. They are born together at $k = 0$.
- **Transcritical:** $\dot{x} = kx - x^2 = x(k - x)$, stationary points 0 and k, $f'(x) = k - 2x$. Then $f'(0) = k$ and $f'(k) = -k$: for $k < 0$, 0 is stable and k unstable; for $k > 0$ the roles swap. The two points exchange stability as they cross.

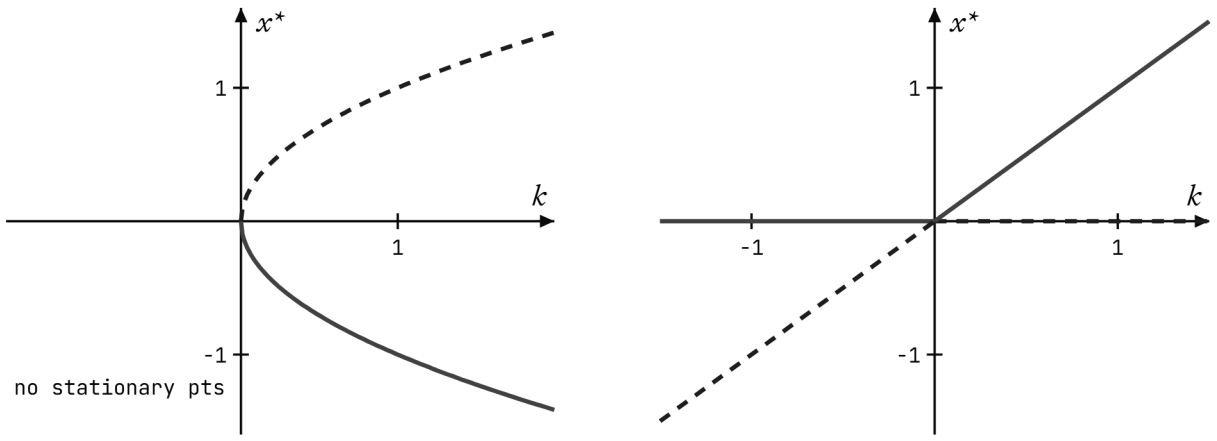


Figure 2.8. *Left: saddle-node bifurcation of $\dot{x} = x^2 - k$ (Robinson Exercise 7.5): two stationary points collide and disappear as k decreases through 0. Right: transcritical bifurcation of $\dot{x} = kx - x^2$ (Exercise 7.6): the branches $x^* = 0$ and $x^* = k$ cross and exchange stability. Green solid: stable; red dashed: unstable.*

SECTION 06 • NOTES P. 3 • ROBINSON SEC. 8.1, 8.4–8.6 • LECTURE SEP 10

6. Explicit solutions (I): separable equations

The second half of p. 3 of the class notes is headed "Explicit solutions to *some* first-order ODEs": three methods that produce formulas, each for a particular class of equations. The first is separation of variables (the notes refer to Sheet 2, Exercises 4–6).

SEPARABLE EQUATIONS [NOTES, P. 3]

$$\begin{cases} x'(t) = f\big(x(t)\big)\, g(t), \\ x(t_0) = x_0. \end{cases}$$

If $f \neq 0$ (never zero), divide both sides by f, separating x and t, and integrate from t_0 to t:

$$\int_{t_0}^{t} \frac{x'(\tau)}{f\big(x(\tau)\big)}\, d\tau = \int_{t_0}^{t} g(\tau)\, d\tau.$$

On the left substitute $u = x(\tau)$, $du = x'(\tau)\, d\tau$; the limits $\tau = t_0$ and $\tau = t$ become $u = x_0$ and $u = x(t)$:

$$\int_{x_0}^{x(t)} \frac{du}{f(u)} = \int_{t_0}^{t} g(\tau)\, d\tau.$$

This is Robinson's formula (8.3); his definition and recipe follow.

DEFINITION — SEPARABLE EQUATION [ROBINSON, EQ. (8.1), P. 59]

$$\frac{dx}{dt} = f(x)\, g(t). \tag{8.1}$$

The right-hand side is a product of a function of x only and a function of t only. Autonomous equations ($g \equiv 1$) and "trivial" equations $\dot{x} = g(t)$ ($f \equiv 1$) are special cases.

6.1 Why dividing by f is legitimate

"If $f \neq 0$ (never)": by Picard's theorem a solution that touches a zero x^* of f at one time is the constant $x \equiv x^*$ for all time. So either $f\big(x(t)\big) = 0$ for every t (a constant solution, to be listed separately) or $f\big(x(t)\big) \neq 0$ for every t, and the division is allowed. Robinson's proof that the recipe is correct:

1. **Constant solutions first.** If $f(x(s)) = 0$ at some time s, then $x(t) \equiv x(s)$ solves the equation ($\dot{x} = 0 = f(x(s))g(t)$), and by uniqueness it is the solution. So either $f(x(t)) = 0$ for every t, or $f(x(t)) \neq 0$ for every t. These constant solutions are lost when dividing by f and must be listed separately.

2. In the second case divide: $\dfrac{1}{f(x)}\dfrac{dx}{dt} = g(t)$ (8.8).

3. Let H be an antiderivative of $1/f$, $H'(x) = 1/f(x)$. By the chain rule

$$\frac{d}{dt}H\big(x(t)\big) = H'\big(x(t)\big)\,\frac{dx}{dt} = \frac{1}{f(x)}\frac{dx}{dt} = g(t).$$

4. Let G be an antiderivative of g. Two functions with the same derivative differ by a constant: $H(x(t)) = G(t) + c$ (8.9). At $t = t_0$: $c = H(x_0) - G(t_0)$, so $H(x(t)) - H(x_0) = G(t) - G(t_0)$, which by Barrow's rule is exactly (8.3) = (8.11).

No infinitesimals were manipulated: the "recipe" is the chain rule read backwards. To obtain $x(t)$ explicitly one must invert H; sometimes the implicit relation $H(x) = G(t) + c$ is the best available.

6.2 First examples

EXAMPLE OF THE NOTES: $x'(t) = kx(t)$ [NOTES, P. 3]

Here $f(x) = x$, $g(t) = k$. Besides the constant solution $x \equiv 0$, for $x \neq 0$: $\int \frac{dx}{x} = \int k\,dt$, so $\ln|x| = kt + c$ and $|x| = e^c e^{kt}$. Since x keeps its sign, $x(t) = Ce^{kt}$ with $C = \pm e^c$ (and $C = 0$ for the constant solution): $x(t) = Ce^{kt}$, as found in Topic 1, Section 5.2.

EXAMPLE 8.1 (ROBINSON P. 60): $\dot{x} = x^2$, $x(0) = x_0$

If $x_0 = 0$, $x \equiv 0$. Otherwise separate, $x^{-2}dx = dt$, and integrate between corresponding limits. An antiderivative of x^{-2} is $-x^{-1}$:

$$\int_{x_0}^{x(t)} \frac{dx}{x^2} = \int_0^t d\tilde{t} \implies \Big[-\frac{1}{x}\Big]_{x_0}^{x(t)} = t \implies -\frac{1}{x(t)} + \frac{1}{x_0} = t \implies x(t) = \frac{1}{x_0^{-1} - t}.$$

This is the blow-up solution (6.7) of Section 3.2, now derived rather than guessed.

6.3 The logistic equation solved

Solve $x'(t) = kx\left(1 - \dfrac{x(t)}{M}\right)$, $x(t_0) = x_0$, with $k, M > 0$ [Notes, p. 3]. (The notes write the initial condition as $x(t_0) = t_0$; $x(t_0) = x_0$ is meant, as the rest of the computation shows.)

1. **Constant solutions first.** "We know from Picard's theorem that x is either 0 or M, or never": if $x_0 = 0$ or $x_0 = M$ the solution is constant, and otherwise $x(t)$ never takes the values 0 or M. Take $0 < x_0 < M$ or $x_0 > M$; then $x(t)$ stays positive and $M - x(t)$ keeps the sign of $M - x_0$.

2. **Separate.** $f(x) = x\left(1 - \frac{x}{M}\right)$, $g(t) = k$:

$$\int_{x_0}^{x} \frac{dx}{x\left(1 - \frac{x}{M}\right)} = \int_{t_0}^{t} k\,dt = k(t - t_0).$$

3 **Partial fractions.** Look for $\dfrac{1}{x\left(1-\frac{x}{M}\right)} = \dfrac{A}{x} + \dfrac{B}{1-\frac{x}{M}}$, i.e. $1 = A\left(1-\frac{x}{M}\right) + Bx$ for all x. At $x = 0$: $1 = A$. At $x = M$: $1 = 0 + BM$, so $B = 1/M$. Hence

$$\frac{1}{x\left(1-\frac{x}{M}\right)} = \frac{1}{x} + \frac{1/M}{1 - x/M} = \frac{1}{x} + \frac{1}{M-x},$$

multiplying top and bottom of the second fraction by M.

4 **Integrate each term.** An antiderivative of $1/x$ ($x > 0$) is $\ln x$. For the second, substitute $u = M - x$, $du = -dx$: $\int \frac{dx}{M-x} = -\int \frac{du}{u} = -\ln|M-x|$. By Barrow's rule,

$$\int_{x_0}^{x} \frac{dx}{x} + \int_{x_0}^{x} \frac{dx}{M-x} = \ln\frac{x}{x_0} + \ln\frac{|M-x_0|}{|M-x|} = \ln\left(\frac{x\,(M-x_0)}{x_0\,(M-x)}\right),$$

where the moduli cancel because $M - x$ and $M - x_0$ have the same sign (step 1).

5 **Exponentiate.** $\ln\left(\dfrac{x(M-x_0)}{x_0(M-x)}\right) = k(t-t_0)$ gives

$$\frac{x}{M-x} = \frac{x_0\,e^{k(t-t_0)}}{M-x_0} =: E(t).$$

6 **Solve for x.** $x = (M-x)E = ME - xE$, so $x\,(1+E) = ME$ and

$$x = \frac{ME}{1+E} = \left(\frac{Mx_0e^{k(t-t_0)}}{M-x_0}\right) \Bigg/ \left(1 + \frac{x_0e^{k(t-t_0)}}{M-x_0}\right) = \frac{Mx_0\,e^{k(t-t_0)}}{M - x_0 + x_0\,e^{k(t-t_0)}}.$$

7 **Divide top and bottom by $e^{k(t-t_0)}$.**

$$x(t) = \frac{Mx_0}{x_0 + (M-x_0)\,e^{-k(t-t_0)}}.$$

LOGISTIC SOLUTION

$$x(t) = \frac{Mx_0}{x_0 + (M-x_0)\,e^{-k(t-t_0)}}, \qquad \lim_{t\to\infty} x(t) = \frac{Mx_0}{x_0+0} = M \quad (x_0 > 0).$$

- **Check at $t = t_0$:** $x = Mx_0/(x_0 + M - x_0) = x_0$.

- **Check the constant solutions:** $x_0 = M$ gives $x \equiv M^2/M = M$; $x_0 = 0$ gives $x \equiv 0$.
- **Small population:** while $x_0 e^{k(t-t_0)} \ll M$, the denominator of the step-6 form is $\approx M$ and $x \approx x_0 e^{k(t-t_0)}$: exponential growth .
- **Inflection:** $x'' = \frac{d}{dt}\left[kx - \frac{k}{M}x^2\right] = \left(k - \frac{2k}{M}x\right)x'$ by the chain rule, so the growth is fastest at $x = M/2$.
- **Dimensions:** $[k(t-t_0)] = \mathrm{s}^{-1} \cdot \mathrm{s} = 1$, and x has the units of M .
- **Robinson's form:** with p for the population, Robinson separates $\frac{M\,dp}{p(M-p)} = k\,dt$ and uses $\frac{M}{p(M-p)} = \frac{1}{p} + \frac{1}{M-p}$ (Eq. 8.13): the same identity as step 3, multiplied by M. He obtains the same formula [Robinson, pp. 66-67].

CENSUS FIT OF THE LOGISTIC MODEL (ROBINSON PP. 67-68)

With $\alpha = e^{50k}$, $p_0 = p(1801)$, $p_1 = p(1851)$, $p_2 = p(1901)$, the solution requires $p_1 = \dfrac{Mp_0\alpha}{M - p_0 + p_0\alpha}$ and $p_2 = \dfrac{Mp_0\alpha^2}{M - p_0 + p_0\alpha^2}$. Eliminating M gives

$$\alpha = \frac{p_2(p_1 - p_0)}{p_0(p_2 - p_1)} = 2.0234, \qquad M = \frac{p_1(2p_0p_2 - p_1p_2 - p_0p_1)}{p_0p_2 - p_1^2} = 83.1 \text{ million}, \tag{8.14}$$

so $k = \ln(2.0234)/50 = 0.0141\ \mathrm{yr}^{-1}$. The model then predicts 66.8 million for 2001, against about 63 million observed – far better than the exponential model's 131.6 million (values recomputed from the census data; they agree with Robinson's).

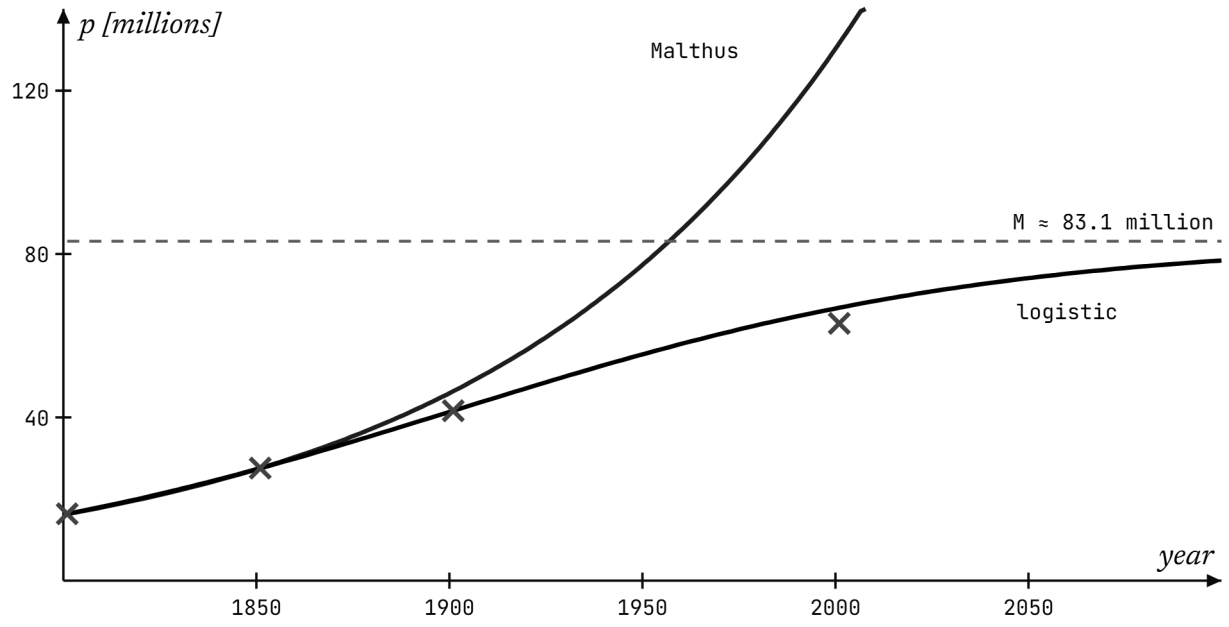


Figure 2.9. *Census data for Great Britain and Ireland (green crosses: 1801, 1851, 1901 and ≈ 63 million in 2001) with the Malthus fit $k \approx 0.0104$ (red) and the logistic fit $k \approx 0.0141$, $M \approx 83.1$ million (indigo). Recomputed from the data of Robinson (8.6), p. 63; compare Figs. 8.2 and 8.4.*

REMOVING MODULUS SIGNS WITH THE PHASE DIAGRAM

Integrals of $1/x$ produce $\ln|x|$. Robinson's advice is to remove the modulus with the phase diagram: a solution cannot cross a stationary point, so $x(t)$ keeps the sign of x_0 (for $\dot{x}=kx$) or of $M-x_0$ (for the logistic equation), as in step 1 above.

6.4 Complement (Robinson): partial fractions for the pitchfork equation

Solve $\dot{x}=x(\kappa^2-x^2)$, $x(0)=x_0$ (the pitchfork with $k=\kappa^2>0$). Separating,

$$\frac{dx}{x(\kappa^2-x^2)}=dt, \qquad \frac{1}{x(\kappa-x)(\kappa+x)}=\frac{1}{\kappa^2}\left[\frac{1}{x}+\frac{1}{2(\kappa-x)}-\frac{1}{2(\kappa+x)}\right].$$

(Check by putting the bracket over the common denominator $2x(\kappa-x)(\kappa+x)$: the numerator is $2(\kappa^2-x^2)+x(\kappa+x)-x(\kappa-x)=2\kappa^2-2x^2+2x^2=2\kappa^2$.) Integrate from 0 to t; with $\int\frac{dx}{\kappa-x}=-\ln|\kappa-x|$ (substitution $u=\kappa-x$):

$$\Big[\ln|x|-\tfrac{1}{2}\ln|\kappa-x|-\tfrac{1}{2}\ln|\kappa+x|\Big]_{x_0}^{x(t)}=\kappa^2 t \implies \ln\frac{|x(t)|\sqrt{|\kappa^2-x_0^2|}}{|x_0|\sqrt{|\kappa^2-x(t)^2|}}=\kappa^2 t.$$

Exponentiate and square: $x(t)^2|\kappa^2-x_0^2|=x_0^2|\kappa^2-x(t)^2|e^{2\kappa^2 t}$ (8.15). The phase diagram shows that $x^2-\kappa^2$ keeps its sign, so the moduli go, and solving the linear equation for $x(t)^2$:

$$x(t)=\pm\sqrt{\frac{\kappa^2}{1+e^{-2\kappa^2 t}\left(\kappa^2 x_0^{-2}-1\right)}}, \tag{8.16}$$

with the sign of x_0. As $t\to\infty$, $e^{-2\kappa^2 t}\to 0$ and $x\to\pm\kappa$: the two stable branches of the pitchfork.

7. Explicit solutions (2): the integrating factor

7.1 Deriving the integrating factor

USING INTEGRATING FACTORS [NOTES, P. 3]

$$x'(t) + p(t)\,x(t) = q(t)$$

In general this is not separable. Multiply the equation by an unknown factor $I(t)$ and choose I so that the left-hand side becomes the derivative $\big(I(t)\,x(t)\big)'$.

1. **Multiply.** $I(t)\,x'(t) + I(t)\,p(t)\,x(t) = I(t)\,q(t)$.

2. **Product rule.** $\big(I(t)\,x(t)\big)' = I'(t)\,x(t) + I(t)\,x'(t)$. For this to equal the left-hand side of step 1, the terms $I\,x'$ cancel and we are left with

$$I'(t)\,x(t) = I(t)\,p(t)\,x(t) \quad\Longrightarrow\quad I'(t) = I(t)\,p(t).$$

3. **Solve for I.** With $I > 0$, divide: $\dfrac{I'(t)}{I(t)} = p(t)$. The left side is $\big(\ln I(t)\big)'$ by the chain rule, so

$$\ln I(t) = \int p(t)\,dt = P(t), \qquad I(t) = e^{P(t)},$$

where P is a primitive of p (for example, $p(t) = t$ gives $P(t) = t^2/2$). This I is the **integrating factor**.

4. **Integrate.** Step 1 now reads $\big(I(t)\,x(t)\big)' = I(t)\,q(t)$. Integrating both sides, $I(t)\,x(t) = \int I(t)\,q(t)\,dt$ (the arbitrary constant is inside the indefinite integral), and dividing by $I(t)$:

GENERAL SOLUTION [NOTES, P. 3]

$$x(t) = \frac{1}{I(t)} \int I(t)\,q(t)\,dt, \qquad I(t) = e^{P(t)}, \quad P' = p.$$

7.2 The initial value problem in three steps

Now consider the IVP [Notes, p. 3]

$$\begin{cases} x'(t) + p(t)\,x(t) = q(t), \quad t > t_0, \\ x(t_0) = x_0. \end{cases}$$

1 **Step 1: find $I(t)$.** Define $P(t) = \displaystyle\int_{t_0}^{t} p(\tau)\,d\tau$. Notice $P(t_0) = \displaystyle\int_{t_0}^{t_0} p(\tau)\,d\tau = 0$, therefore $I(t) = e^{P(t)}$ satisfies $I(t_0) = e^0 = 1$.

2 **Step 2: integrate from t_0 to t.** With the knowledge of $I(t)$,

$$\int_{t_0}^{t} \big(I(\tau)\,x(\tau)\big)'\,d\tau = \int_{t_0}^{t} I(\tau)\,q(\tau)\,d\tau.$$

3 **Step 3: simplify.** By Barrow's rule the left side is $I(t)\,x(t) - I(t_0)\,x(t_0) = I(t)\,x(t) - 1 \cdot x_0$, so $I(t)\,x(t) = x_0 + \int_{t_0}^{t} I(\tau)\,q(\tau)\,d\tau$. Divide by $I(t)$.

SOLUTION OF THE IVP [NOTES, P. 3]

$$x(t) = \frac{x_0}{I(t)} + \frac{1}{I(t)} \int_{t_0}^{t} I(\tau)\,q(\tau)\,d\tau, \qquad I(t) = e^{P(t)}, \quad P(t) = \int_{t_0}^{t} p(\tau)\,d\tau.$$

The first term is the **free response** (the initial value carried by the homogeneous equation, $q = 0$); the second is the **forced response** to the input q. The same "homogeneous + particular" structure reappears in Topic 3. Robinson writes the formula with any primitive P, which adds the factor $e^{P(t_0)}$ to x_0; choosing $P(t_0) = 0$, as the notes do, removes it.

WORKED EXAMPLE (WRITTEN FOR THIS PAGE): $x' + t\,x = t$, $x(0) = x_0$

This uses the primitive mentioned in the notes, $p(t) = t$. Here $q(t) = t$ and $t_0 = 0$.

1 **Step 1.** $P(t) = \int_0^t \tau\,d\tau = \left[\frac{\tau^2}{2}\right]_0^t = \frac{t^2}{2}$, so $I(t) = e^{t^2/2}$ and $I(0) = 1$.

2 **Step 2.** $\displaystyle\int_0^t I(\tau)\,q(\tau)\,d\tau = \int_0^t \tau\,e^{\tau^2/2}\,d\tau$. Substitute $u = \tau^2/2$, $du = \tau\,d\tau$; the limits become $u = 0$ and $u = t^2/2$:

$$\int_0^{t^2/2} e^u\,du = \left[e^u\right]_0^{t^2/2} = e^{t^2/2} - 1.$$

3 **Step 3.** $x(t) = x_0 e^{-t^2/2} + e^{-t^2/2}\left(e^{t^2/2} - 1\right) = 1 + (x_0 - 1)\,e^{-t^2/2}$.

Check: $x' = (x_0 - 1)\,e^{-t^2/2}\cdot(-t)$ by the chain rule, so $x' + tx = -t(x_0-1)e^{-t^2/2} + t + t(x_0 - 1)e^{-t^2/2} = t$, and $x(0) = 1 + x_0 - 1 = x_0$. As $t \to \infty$, $x \to 1$ for every x_0. (The equation is also separable, $x' = t(1-x)$, which gives the same answer.)

7.3 Complement (Robinson): the idea with constant coefficients

The general linear first-order equation is $a_1(t)\dot{x} + a_0(t)x = f(t)$. Where $a_1 \neq 0$ we divide by it and obtain the standard form [Robinson, Eq. (9.1), p. 75]

$$\frac{dx}{dt} + p(t)\,x = q(t). \tag{9.1}$$

It is separable when p and q are constants (and in a few special cases such as $q \equiv 0$); the integrating factor works for all p, q.

For $\dot{x} + px = q$ with constants p, q, notice that by the product rule

$$\frac{d}{dt}\big(x\,e^{pt}\big) = \dot{x}\,e^{pt} + x\,p\,e^{pt} = e^{pt}\big(\dot{x} + px\big). \tag{9.3}$$

So multiplying the equation by e^{pt} turns its left side into an exact derivative:

$$\frac{d}{dt}\big(xe^{pt}\big) = qe^{pt} \implies xe^{pt} = \frac{q}{p}e^{pt} + C \implies x(t) = \frac{q}{p} + Ce^{-pt}. \tag{9.5}$$

If $p > 0$, every solution tends to q/p as $t \to \infty$, whatever the initial condition. With $x(a) = x_a$: $C = (x_a - q/p)e^{pa}$ and $x(t) = q/p + (x_a - q/p)e^{-p(t-a)}$ (9.6).

Robinson's worked examples follow.

EXAMPLE 9.1 (ROBINSON PP. 78–79): $\dot{x} + 3x = t$, $x(0) = 8/9$

1. $p = 3$, so $I = e^{\int 3\,dt} = e^{3t}$ and $\dfrac{d}{dt}\big(xe^{3t}\big) = te^{3t}$.
2. Antiderivative of se^{3s} by parts, with $u = s$, $dv = e^{3s}ds$, $du = ds$, $v = e^{3s}/3$: $\displaystyle\int se^{3s}ds = \frac{se^{3s}}{3} - \int \frac{e^{3s}}{3}ds = \frac{se^{3s}}{3} - \frac{e^{3s}}{9}$.
3. Integrate from 0 to t (Barrow):

$$x(t)e^{3t} - x(0) = \Big[\frac{se^{3s}}{3} - \frac{e^{3s}}{9}\Big]_0^t = \frac{te^{3t}}{3} - \frac{e^{3t}}{9} - \Big(0 - \frac{1}{9}\Big).$$

4. With $x(0) = 8/9$: $xe^{3t} = \dfrac{te^{3t}}{3} - \dfrac{e^{3t}}{9} + 1$, so $x(t) = e^{-3t} + \dfrac{t}{3} - \dfrac{1}{9}$.

Check: $\dot{x}+3x=(-3e^{-3t}+\tfrac{1}{3})+(3e^{-3t}+t-\tfrac{1}{3})=t$; $x(0)=1-\tfrac{1}{9}=\tfrac{8}{9}$.

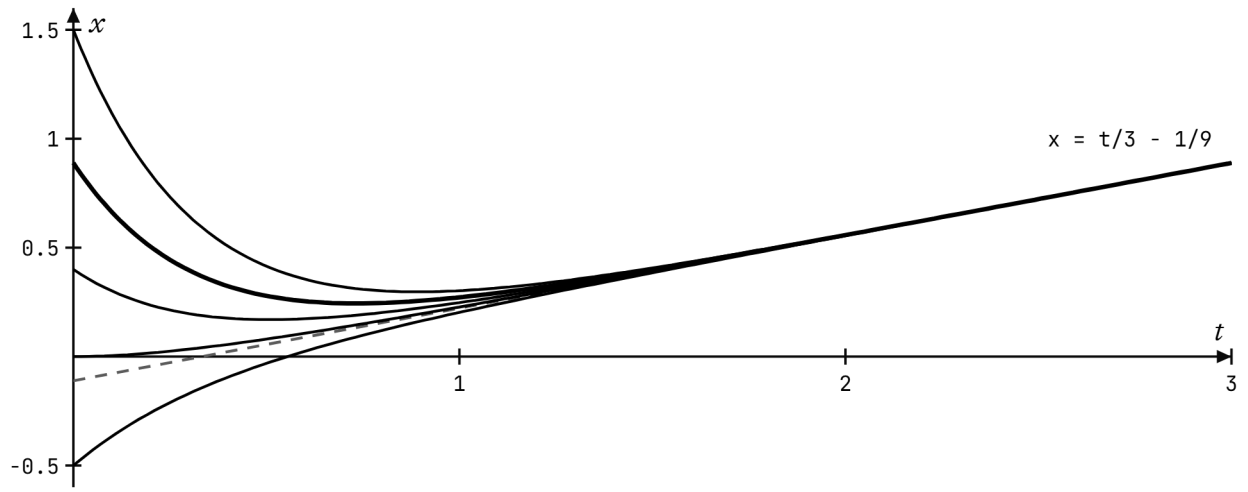


Figure 2.10. *Solutions of $\dot{x}+3x=t$ (Robinson Example 9.1) for several $x(0)$; the thick curve is $x(0)=8/9$. Every solution approaches the line $x=t/3-1/9$ because the term Ce^{-3t} decays.*

EXAMPLE 9.2 (ROBINSON P. 79): $(x^2+1)\,y'+4xy=12x$

1. Standard form (divide by $x^2+1>0$): $y'+\dfrac{4x}{x^2+1}y=\dfrac{12x}{x^2+1}$.
2. Integrating factor: substitute $u=x^2+1$, $du=2x\,dx$: $\displaystyle\int\frac{4x}{x^2+1}dx=\int\frac{2\,du}{u}=2\ln u$, so $I=e^{2\ln(x^2+1)}=(x^2+1)^2$.
3. $\dfrac{d}{dx}\big[y(x^2+1)^2\big]=12x(x^2+1)$. With the same substitution, $\int 12x(x^2+1)\,dx=\int 6u\,du=3u^2=3(x^2+1)^2$.
4. $y(x^2+1)^2=3(x^2+1)^2+c$, so $\;y(x)=3+\dfrac{c}{(x^2+1)^2}$.

7.4 Complement (Robinson): Newton's law of cooling

An object at temperature $T(t)$ in surroundings at temperature $A(t)$ exchanges heat at a rate proportional to the temperature difference:

$$\frac{dT}{dt}=-k\big(T-A(t)\big),\qquad k>0\ [\mathrm{s}^{-1}]. \tag{9.9}$$

In standard form, $\dot{T}+kT=kA(t)$, with integrating factor e^{kt}.

CONSTANT SURROUNDINGS: ESTIMATING THE TIME OF DEATH (SEC. 9.4.1, PP. 80–81)

With A constant, integrating $\frac{d}{dt}[Te^{kt}]=kAe^{kt}$ between t_1 and t_2 gives $T(t_2)e^{kt_2}-T(t_1)e^{kt_1}=A\big(e^{kt_2}-e^{kt_1}\big)$, i.e.

$$T(t_2) = A + \big[T(t_1) - A\big] e^{-k(t_2 - t_1)}.$$

Data (time in hours): room at $A = 24\,^\circ\mathrm{C}$, body at $28\,^\circ\mathrm{C}$ at 8 a.m. and $26\,^\circ\mathrm{C}$ at 9 a.m. Then $26 = 24 + 4e^{-k}$, so $e^{-k} = 1/2$ and $k = \ln 2\ \mathrm{h}^{-1}$. The time of death t_0 satisfies $T(t_0) = 37$: $28 = 24 + 13e^{-k(8 - t_0)}$, so $4 = 13e^{-k(8-t_0)}$ and

$$8 - t_0 = \frac{\ln 4 - \ln 13}{-\ln 2} = \frac{\ln(13/4)}{\ln 2} = \frac{1.1787}{0.6931} = 1.70\ \mathrm{h}.$$

ERRATUM — ROBINSON P. 81

The book's equation for $8 - t_0$ is correct, but it then states "$t_0 \approx 1.7$, putting the time of death at approximately 1:42 a.m.". The value 1.7 is $8 - t_0$, not t_0. The correct result is $t_0 = 8 - 1.70 = 6.30$, i.e. about **6:18 a.m.**

Check by substitution. With $t_0 = 6.30$: $T(8) = 24 + 13 \cdot 2^{-1.70} = 24 + 13 \times 0.308 = 28.0\,^\circ\mathrm{C}$. With the book's $t_0 = 1.7$: $T(8) = 24 + 13 \cdot 2^{-6.3} = 24 + 13 \times 0.0127 = 24.2\,^\circ\mathrm{C} \neq 28$. Physically, a body that cools by $2\,^\circ\mathrm{C}$ per hour near $28\,^\circ\mathrm{C}$ cannot have been at $37\,^\circ\mathrm{C}$ more than six hours earlier.

Periodic surroundings: amplitude attenuation and lag (Sec. 9.4.2–9.4.4)

Take $A(t) = \mu + a\cos\omega t$ (mean μ, amplitude a, period $2\pi/\omega$). Integrating (9.9) from 0 to t gives $T(t)e^{kt} - T(0) = k\int_0^t A(\tilde t)e^{k\tilde t}d\tilde t$ (9.10). Using the antiderivative (9.11) $\int e^{kt}\cos\omega t\,dt = \dfrac{e^{kt}(k\cos\omega t + \omega\sin\omega t)}{k^2+\omega^2}$ (verify by differentiating) and combining $A\cos\omega t + B\sin\omega t = \sqrt{A^2+B^2}\cos(\omega t - \phi)$, $\tan\phi = B/A$ (9.12)–(9.15), Robinson obtains

$$T(t) = \mu + \left(T(0) - \mu - \frac{ak^2}{k^2+\omega^2}\right)e^{-kt} + \frac{ak}{\sqrt{k^2+\omega^2}}\cos(\omega t - \phi), \qquad \phi = \tan^{-1}\frac{\omega}{k}. \tag{9.17}$$

- **Transient:** the middle term decays on the time scale $1/k$ and the memory of $T(0)$ is lost.
- **Attenuation:** the inside oscillation has amplitude $a\,k/\sqrt{k^2+\omega^2} < a$ (9.18).
- **Lag:** it peaks a time $\phi/\omega = \omega^{-1}\tan^{-1}(\omega/k)$ after the outside (9.19).
- **Limits:** $k \gg \omega$ (good thermal contact): amplitude $\to a$, lag $\to 0$, the object follows its surroundings. $k \ll \omega$ (heavy insulation): amplitude $\approx ak/\omega \to 0$, lag $\to \pi/(2\omega)$, a quarter period.

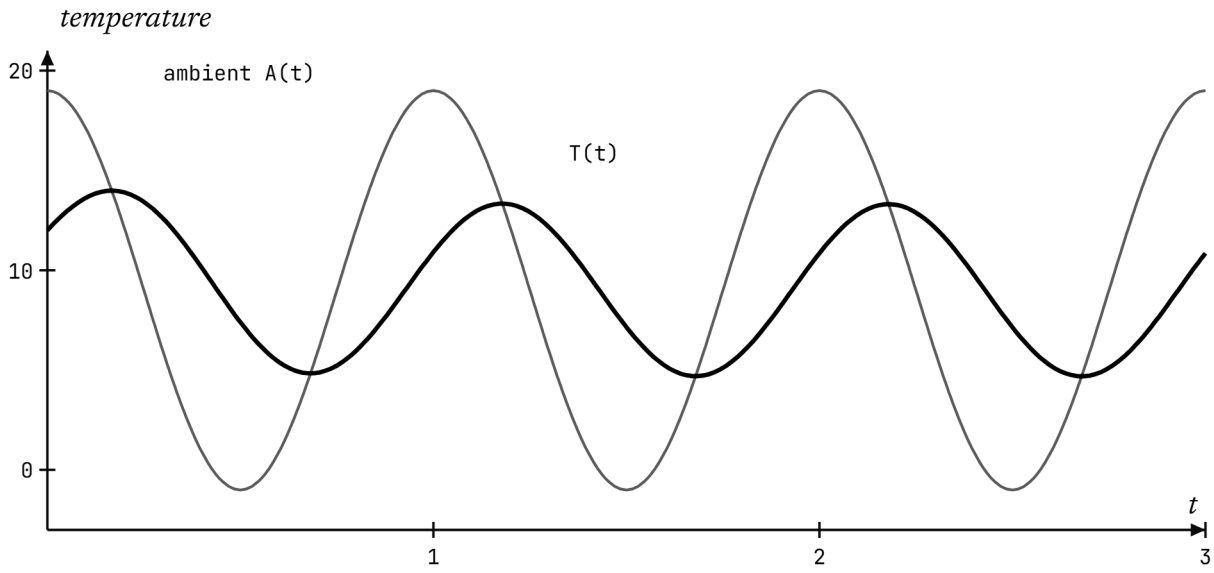


Figure 2.11. *Newton's law of cooling with a periodic surrounding temperature $A(t) = 9 + 10\cos 2\pi t$, $k = 3$, $T(0) = 12$ (data of Robinson Fig. 9.3). After the transient dies out, T oscillates with the same period, a smaller amplitude and a time lag.*

AEROSPACE MODEL — CABIN VENTILATION AS A STIRRED TANK (COMPLEMENTARY, BEYOND THE REFERENCE TEXT)

A well-mixed volume V [m³] (an aircraft cabin, a combustor, a continuously stirred tank) receives a volumetric flow Q [m³/s] of air with contaminant concentration c_{in}, and the same flow leaves with the cabin concentration $c(t)$ (perfect mixing). Occupants add a source S [kg/s]. The mass balance "rate of change = in − out + source" reads

$$\frac{d}{dt}\big(Vc\big) = Qc_{\text{in}} - Qc + S \quad\Longrightarrow\quad \frac{dc}{dt} + \frac{Q}{V}\,c = \frac{Qc_{\text{in}} + S}{V}.$$

This is (9.1) with constant $p = Q/V$ and $q = (Qc_{\text{in}} + S)/V$, so by (9.6) $c(t) = c_\infty + (c_0 - c_\infty)e^{-t/\tau}$, with steady state $c_\infty = c_{\text{in}} + S/Q$ and **residence time** $\tau = V/Q$. For a cabin of $V = 150\ \text{m}^3$ supplied with $Q = 0.5\ \text{m}^3/\text{s}$, $\tau = 300\ \text{s}$: after a disturbance the air is 95% refreshed in $3\tau = 15$ minutes ($e^{-3} \approx 0.05$). Dimensions: $[Q/V] = \text{s}^{-1}$, $[S/Q] = \text{kg/m}^3$. Newton's law of cooling is the same equation with $k = Q/V$.

8. Explicit solutions (3): exact equations

8.1 From a family of curves to an ODE

EXACT EQUATIONS [NOTES, P. 4]

Let y be a function of x, and consider the function $F(x, y) = C$, where C is a constant. Take the derivative with respect to x:

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\,\frac{dy}{dx} = 0,$$

which is a first-order ODE. Therefore, if we have an ODE of the form

$$f(x, y) + g(x, y)\,\frac{dy}{dx} = 0$$

and we find $F(x, y)$ such that $\dfrac{\partial F}{\partial x} = f$ and $\dfrac{\partial F}{\partial y} = g$, then the solution to the ODE is $F(x, y) = C$.

The derivative in the box is the multivariable chain rule applied to $x \mapsto F\big(x, y(x)\big)$: the first argument changes at rate 1 and the second at rate dy/dx, so $\frac{d}{dx}F\big(x, y(x)\big) = F_x \cdot 1 + F_y \cdot \frac{dy}{dx}$. A constant has zero derivative.

An observation. If such an F exists, then

$$\frac{\partial f}{\partial y} = \frac{\partial^2 F}{\partial y\,\partial x} = \frac{\partial^2 F}{\partial x\,\partial y} = \frac{\partial g}{\partial x},$$

because the mixed second partial derivatives of a function with continuous second derivatives are equal (Schwarz's theorem). So the condition "there is an F" can be replaced by a condition on f and g alone [Notes, p. 4].

THEOREM [NOTES, P. 4]

Let $f, g, \dfrac{\partial f}{\partial y}$ and $\dfrac{\partial g}{\partial x}$ be continuous functions. Then $f(x, y) + g(x, y)\dfrac{dy}{dx} = 0$ is an **exact** ODE if

$$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x},$$

and its solution is $F(x,y)=C$, with $\dfrac{\partial F}{\partial x}=f$ and $\dfrac{\partial F}{\partial y}=g$.

Complement (Robinson, Sec. 10.1). The condition is necessary and, on a rectangle (more generally a simply connected domain), also sufficient: the equation is exact *if and only if* $f_y=g_x$ there. The observation proves the "only if" part. Robinson's proof of the "if" part constructs F, and is the method used in the examples:

PROOF

Necessity. If $f=F_x$ and $g=F_y$, then $f_y=F_{xy}$ and $g_x=F_{yx}$, and mixed second partial derivatives of a smooth function are equal.

Sufficiency (constructive). Define F by integrating f in x, holding y fixed, with a "constant" that may depend on y:

$$F(x,y)=\int f(x,y)\,dx+C(y). \tag{10.6}$$

Then $F_x=f$ automatically. We need $F_y=g$:

$$\frac{\partial}{\partial y}\int f\,dx+C'(y)=g \quad\Longrightarrow\quad C'(y)=g(x,y)-\frac{\partial}{\partial y}\int f(x,y)\,dx. \tag{10.7}$$

This defines $C(y)$ only if the right side does not depend on x. Differentiate it in x:

$$\frac{\partial}{\partial x}\left[g-\frac{\partial}{\partial y}\int f\,dx\right]=\frac{\partial g}{\partial x}-\frac{\partial}{\partial y}\frac{\partial}{\partial x}\int f\,dx=\frac{\partial g}{\partial x}-\frac{\partial f}{\partial y}=0$$

by the condition $\partial f/\partial y=\partial g/\partial x$. So the right side of (10.7) is a function of y alone, and integrating it gives $C(y)$.

■

8.2 Example 1 of the notes

EX 1: $x^3+\dfrac{y}{x}+\left(y^2+\ln x\right)\dfrac{dy}{dx}=0,\ x>0$ [NOTES, P. 4; ROBINSON EXAMPLE 10.1, P. 91]

1. $f=x^3+\dfrac{y}{x}, g=y^2+\ln x$. The ODE is **not linear**, because of the term $y^2\dfrac{dy}{dx}$.
2. Test: $\dfrac{\partial f}{\partial y}=\dfrac{1}{x}$ and $\dfrac{\partial g}{\partial x}=\dfrac{1}{x}$. Equal: the equation is exact.

3. Integrate f in x (y held fixed), with a "constant" $\varphi(y)$:

$$F(x,y) = \int \left(x^3 + \frac{y}{x}\right) dx = \frac{x^4}{4} + y\ln x + \varphi(y).$$

4. Impose $\dfrac{\partial F}{\partial y} = g$: $\ln x + \varphi'(y) = y^2 + \ln x$, so $\varphi'(y) = y^2$ (indeed free of x) and $\varphi(y) = \dfrac{y^3}{3} + c$.

5. Solution:

$$F(x,y) = \frac{x^4}{4} + \frac{y^3}{3} + y\ln x = C.$$

It is an **implicit** solution, "since x and y are together" in the words of the notes: it is left in implicit form (isolating y would need Cardano's formula for the cubic $\frac{y^3}{3} + y\ln x = C - \frac{x^4}{4}$).

Check: $F_x + F_y\,y' = \big(x^3 + y/x\big) + \big(y^2 + \ln x\big)y' = f + g\,y'$.

8.3 Example 2 of the notes

EX 2: $\big(1 + e^x y + x e^x y\big) + \big(x e^x + 2\big)\dfrac{dy}{dx} = 0$ [NOTES, P. 4]

1. $f = 1 + e^x y + x e^x y,\; g = x e^x + 2.$

2. Test. $\dfrac{\partial f}{\partial y} = e^x + x e^x$. For g, the product rule on $x e^x$ gives $\dfrac{\partial g}{\partial x} = 1\cdot e^x + x\cdot e^x = e^x + x e^x$. Equal: the equation is exact.

3. Integrate f in x. The term $x e^x y$ needs integration by parts, with $u = x$, $dv = e^x\,dx$, $du = dx$, $v = e^x$:

$$\int x e^x\,dx = x e^x - \int e^x\,dx = x e^x - e^x.$$

Hence

$$F = \int \big(1 + e^x y + x e^x y\big)\,dx = x + y e^x + y\big(x e^x - e^x\big) + \varphi(y) = x + y x e^x + \varphi(y),$$

since $y e^x - y e^x = 0$.

4. Impose $\dfrac{\partial F}{\partial y} = g$: $x e^x + \varphi'(y) = x e^x + 2$, so $\varphi'(y) = 2$ and $\varphi(y) = 2y + c$.

5 Solution: $F(x,y) = x + yxe^x + 2y = C$. Here y appears only to the first power, so it can be isolated: $y\,(xe^x + 2) = C - x$, and

$$y(x) = \frac{C - x}{xe^x + 2}.$$

The denominator never vanishes: $\frac{d}{dx}(xe^x) = (1+x)e^x$ is zero only at $x = -1$, where xe^x has its minimum $-1/e$, so $xe^x + 2 \geq 2 - 1/e > 0$. The solution is defined for every x.

Check: differentiate $y\,(xe^x + 2) = C - x$ with the product rule: $y'(xe^x + 2) + y(e^x + xe^x) = -1$, i.e. $\big(1 + e^x y + xe^x y\big) + \big(xe^x + 2\big)y' = 0$. (This equation is also linear in y. In the normal form $y' + p\,y = q$ of Section 7, $p = \dfrac{(1+x)e^x}{xe^x + 2}$ and $q = \dfrac{-1}{xe^x + 2}$; since $(xe^x + 2)' = (1+x)e^x$, $P = \ln(xe^x + 2)$ and $I = e^P = xe^x + 2$. Multiplying by I gives back $\frac{d}{dx}\big[(xe^x + 2)\,y\big] = -1$, and the same answer follows.)

8.4 Complement (Robinson): integrating factors for non-exact equations

Multiplying by a well-chosen $I(x,y)$ can make an equation exact. The linear equation $y' + p(x)y = 0$, written $p\,y + 1 \cdot y' = 0$, is not exact ($\partial_y(py) = p \neq 0 = \partial_x 1$) but $e^{P(x)}$ makes it exact: $\partial_y\big(pe^P y\big) = pe^P = \partial_x e^P$. So the integrating factor of Section 7 is a special case.

In general, $I f + I g\,y' = 0$ is exact when $\partial_y(If) = \partial_x(Ig)$. By the product rule this is

$$I_y f + I f_y = I_x g + I g_x \quad\Longleftrightarrow\quad \left(\frac{\partial f}{\partial y} - \frac{\partial g}{\partial x}\right) I = g\frac{\partial I}{\partial x} - f\frac{\partial I}{\partial y}, \tag{10.9}$$

a PDE for I that is in general no easier than the original problem. It becomes an ODE if I depends on one variable only.

SPECIAL INTEGRATING FACTORS

$I = I(x)$ (Robinson (10.10)–(10.11)): then $I_y = 0$ and (10.9) becomes

$$\frac{dI}{dx} = \frac{1}{g}\left(\frac{\partial f}{\partial y} - \frac{\partial g}{\partial x}\right) I,$$

which has a solution $I(x)$ provided $\dfrac{1}{g}\,(f_y - g_x)$ depends on x only. Then $I(x) = \exp\displaystyle\int \frac{f_y - g_x}{g}\,dx$.

$I = I(y)$ (the symmetric case, not treated in the book): then $I_x = 0$ and (10.9) becomes

$$\frac{dI}{dy} = \frac{1}{f}\left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) I,$$

which has a solution $I(y)$ provided $\dfrac{1}{f}\,(g_x - f_y)$ depends on y only. Then $I(y) = \exp\displaystyle\int \frac{g_x - f_y}{f}\,dy$.

EXAMPLE 10.2 (ROBINSON PP. 93–94): AN INTEGRATING FACTOR $I(x)$

$$\left(3\sin y + 5ye^{5x} + \frac{2ye^{5x}}{x}\right) + \left(x\cos y + e^{5x}\right)\frac{dy}{dx} = 0.$$

1. $f_y = 3\cos y + 5e^{5x} + \dfrac{2e^{5x}}{x}$ and $g_x = \cos y + 5e^{5x}$ (chain rule on e^{5x}). Not exact.
2. $\dfrac{f_y - g_x}{g} = \dfrac{2\cos y + 2e^{5x}/x}{x\cos y + e^{5x}} = \dfrac{(2/x)\,(x\cos y + e^{5x})}{x\cos y + e^{5x}} = \dfrac{2}{x}$ – a function of x only.
3. $\dfrac{dI}{dx} = \dfrac{2}{x}I$, so $\ln I = 2\ln x$ and $I = x^2$.
4. New equation: $\left(3x^2\sin y + 5x^2ye^{5x} + 2xye^{5x}\right) + \left(x^3\cos y + x^2e^{5x}\right)y' = 0$. Test: both partials equal $3x^2\cos y + 5x^2e^{5x} + 2xe^{5x}$.
5. $F = \int \tilde{g}\,dy = x^3\sin y + x^2ye^{5x} + C(x)$; then $F_x = 3x^2\sin y + 2xye^{5x} + 5x^2ye^{5x} + C'(x) = \tilde{f}$ gives $C' = 0$.
6. Solution: $x^3\sin y + x^2ye^{5x} = c$.

ILLUSTRATION — AN INTEGRATING FACTOR $I(y)$

Solve $y + \left(2x + y^2\right)\dfrac{dy}{dx} = 0, y > 0$.

1. $f = y, g = 2x + y^2$: $f_y = 1, g_x = 2$. Not exact.
2. $\dfrac{f_y - g_x}{g} = \dfrac{-1}{2x + y^2}$ depends on both variables, so no $I(x)$. But $\dfrac{g_x - f_y}{f} = \dfrac{1}{y}$ depends on y only.
3. $\dfrac{dI}{dy} = \dfrac{I}{y}$, so $I(y) = y$.
4. New equation: $y^2 + (2xy + y^3)y' = 0$. Test: $\partial_y(y^2) = 2y = \partial_x(2xy + y^3)$.
5. $F = \int y^2\,dx = xy^2 + C(y)$; $F_y = 2xy + C'(y) = 2xy + y^3$, so $C(y) = y^4/4$.
6. Solution: $xy^2 + \dfrac{y^4}{4} = c$. Check: $\frac{d}{dx}$ gives $y^2 + 2xyy' + y^3y' = y\,[\,y + (2x + y^2)y'\,] = 0$.

SECTION 09 • ROBINSON SEC. 10.2 • COMPLEMENT

9. Complement: homogeneous and Bernoulli substitutions

Robinson's Chapter 10 closes with two substitutions that turn special nonlinear equations into ones already solved. They are not in the lecture map but appear in the problem sheets' style of questions.

- **Homogeneous equations** $\dfrac{dy}{dx} = F\left(\dfrac{y}{x}\right)$ (a different meaning of "homogeneous" from the linear one). Put $y = ux$; by the product rule $y' = u + xu'$, so $x\dfrac{du}{dx} = F(u) - u$, which is separable. Example 10.3: $xyy' = 2x^2 + 3y^2$ gives $y = \pm x\sqrt{\alpha x^4 - 1}$.
- **Bernoulli equations** $\dfrac{dy}{dx} + p(x)y = q(x)y^n$ (10.13), $n \neq 0, 1$. Put $u = y^{1-n}$; by the chain rule $u' = (1-n)y^{-n}y' = (1-n)y^{-n}\big(-py + qy^n\big) = (1-n)(-pu + q)$, so

$$\frac{du}{dx} + (1-n)p(x)\,u = (1-n)q(x),$$

  which is linear and solved by an integrating factor.

EXAMPLE 10.4 (ROBINSON P. 96): $y' - 6xy = 2xy^2$

$n = 2$, $u = y^{-1}$, $u' = -y^{-2}y' = -y^{-2}(6xy + 2xy^2) = -6xu - 2x$, so $u' + 6xu = -2x$. Integrating factor $e^{\int 6x\,dx} = e^{3x^2}$: $\frac{d}{dx}\big(ue^{3x^2}\big) = -2xe^{3x^2}$; with $w = 3x^2$, $dw = 6x\,dx$, $\int -2xe^{3x^2}dx = -\frac{1}{3}e^{3x^2}$. Hence $u = -\frac{1}{3} + ce^{-3x^2}$ and $y = 1/u = \dfrac{3}{Ce^{-3x^2} - 1}$ with $C = 3c$.

SECTION 10 • SUMMARY

10. Choosing a method

IF THE EQUATION IS…	RECOGNISE IT BY	DO THIS	NOTES	ROBINSON
An IVP $y' = f(x,y)$, $y(x_0) = y_0$	$f, \partial f/\partial y$ continuous near (x_0, y_0)	Picard: a unique solution exists for $\lvert x - x_0\rvert < h$.	p. 3	Sec. 6.2
Autonomous $x' = f(x)$	No explicit t	Stationary points $f(x^*) = 0$; stable if $f'(x^*) < 0$, unstable if $f'(x^*) > 0$; phase line.	p. 3	Ch. 7
Separable $x' = f(x)g(t)$	Right side factorises	List constant solutions $f(x^*) = 0$; then $\int_{x_0}^{x} du/f(u) = \int_{t_0}^{t} g\,d\tau$; remove moduli with the phase line.	p. 3	Ch. 8

IF THE EQUATION IS…	RECOGNISE IT BY	DO THIS	NOTES	ROBINSON
Linear $x' + p(t)x = q(t)$	x and x' appear to the first power	$P = \int_{t_0}^{t} p$, $I = e^{P}$; $x = \frac{x_0}{I} + \frac{1}{I}\int_{t_0}^{t} Iq$.	p. 3	Ch. 9
Exact $f + g\,y' = 0$	$f_y = g_x$	Build F with $F_x = f$, $F_y = g$ (integrate f in x, fix $\varphi(y)$); answer $F = C$.	p. 4	Sec. 10.1
Not exact	$(f_y - g_x)/g$ depends on x only, or $(g_x - f_y)/f$ on y only	Multiply by $I(x)$ or $I(y)$, then as above.	–	Sec. 10.1.1
Homogeneous / Bernoulli	$y' = F(y/x)$ / $y' + py = qy^n$	$y = ux$ / $u = y^{1-n}$.	–	Sec. 10.2

ALWAYS CHECK

Substitute the answer back into the equation and the initial condition. Look for lost constant solutions after dividing. Test the limits $t \to 0$ and $t \to \infty$ against the phase line. Check the dimensions of every constant that appears.